

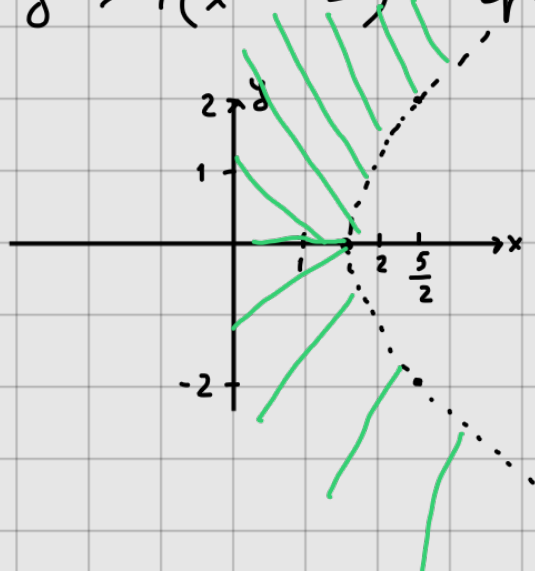
1. Построить область определения функции.

1.1. $z = \ln(y^2 - 4x + 6)$

$D(z): y^2 - 4x + 6 > 0$

$y^2 > 4x - 6$

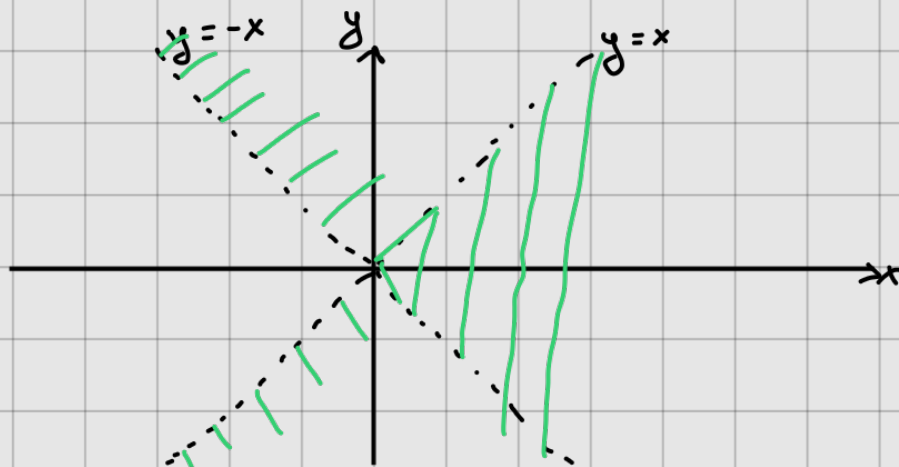
$y^2 > 4(x - \frac{3}{2})$ - граница



1.2.

$z = \frac{1}{\sqrt{x+y}} + \frac{1}{\sqrt{x-y}}$

$D(z): \begin{cases} x+y > 0 \\ x-y > 0 \end{cases} \Leftrightarrow \begin{cases} y > -x \\ y < x \end{cases}$



1.3.

(если $x < 0$, то знаки изменяются)

$$z = \arccos \frac{y-2}{x}$$

$$\begin{cases} y \geq x+2 \\ y \leq 2-x \end{cases} \quad \begin{cases} y \geq 2-x \\ y \leq x+2 \end{cases}$$

$$-1 \leq \frac{y-2}{x} \leq 1$$

$$\begin{cases} \frac{y-2}{x} \geq -1 \\ \frac{y-2}{x} \leq 1 \end{cases} \quad \textcircled{1} \frac{y-2}{x} = -1$$

$$\frac{y-2+x}{x} = 0$$

$$x \neq 0$$

$$y-2+x=0$$

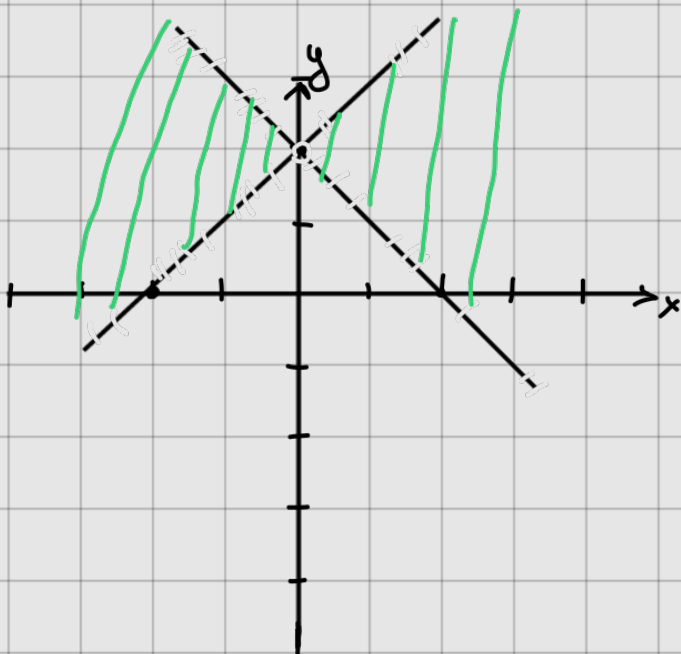
$$y = 2-x$$

$$\textcircled{2} \frac{y-2}{x} = 1$$

$$\frac{y-2-x}{x} = 0$$

$$x \neq 0$$

$$y = 2+x$$



1.4.

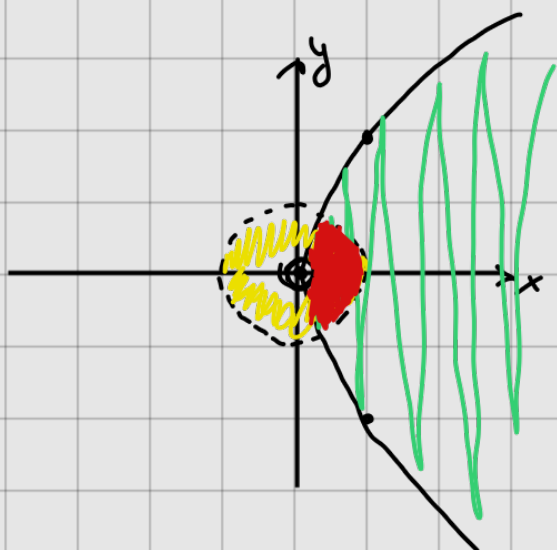
$$z = \sqrt{x - \sqrt{y}}$$

$$\mathcal{D}(z) : \begin{cases} x - \sqrt{y} \geq 0 \\ y \geq 0 \end{cases} \Leftrightarrow \begin{cases} x \geq \sqrt{y} \\ y \geq 0 \end{cases}$$



1.5.

$$z = \frac{\sqrt{4x - y^2}}{\ln(1 - x^2 - y^2)}$$



$$D(z): \begin{cases} 4x - y^2 \geq 0 \\ 1 - x^2 - y^2 > 0 \\ 1 - x^2 - y^2 \neq 1 \end{cases}$$

$$\textcircled{1} y^2 \leq 4x$$

$$\textcircled{2} x^2 + y^2 < 1$$

$$\textcircled{3} x^2 + y^2 \neq 0 \Rightarrow x \neq 0; y \neq 0$$

2. Определить вид поверхности 2-го порядка

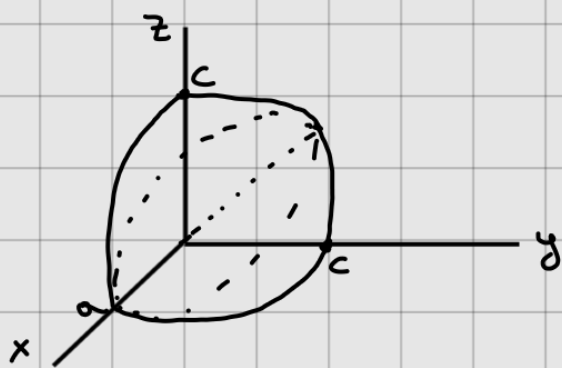
$$2.1. x^2 + 3y^2 + 4z^2 - 5 = 0$$

$$x^2 + 3y^2 + 4z^2 = 5 \quad | :5$$

$$\frac{x^2}{5} + \frac{3}{5}y^2 + \frac{4}{5}z^2 = 1$$

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1 \quad - \text{Эллипсоид}$$

$$a = \sqrt{5} \quad b = \sqrt{\frac{5}{3}} \quad c = \sqrt{\frac{5}{4}}$$



2.2.

вдоль оси Oy

$$3x^2 - 4y^2 + 6z^2 - 6 = 0$$

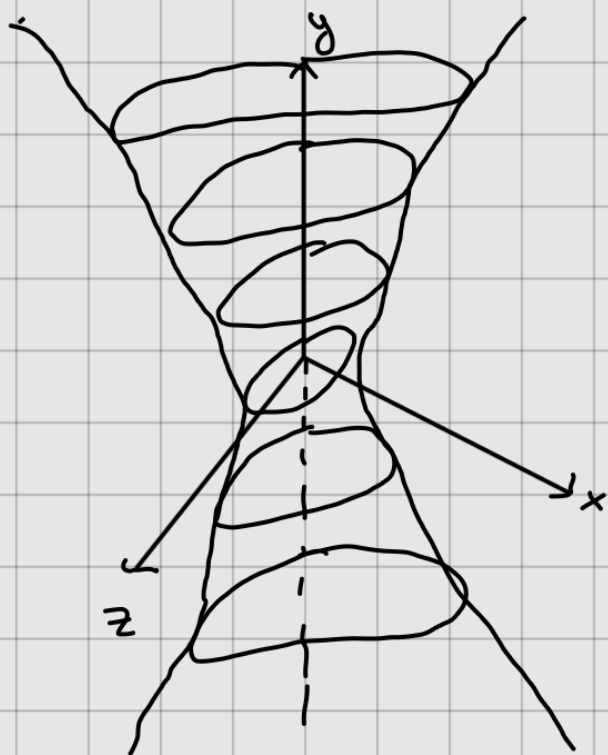
$$3x^2 - 4y^2 + 6z^2 = 6 \quad | :6$$

$$\frac{1}{2}x^2 - \frac{2}{3}y^2 + z^2 = 1$$

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$$

$$a = \sqrt{2}, \quad b = \sqrt{\frac{3}{2}}, \quad c = 1$$

$$\frac{x^2}{a^2} + \frac{z^2}{c^2} = 1 + \frac{y^2}{b^2}$$



Односторонний
гиперболоид
вдоль оси Oy

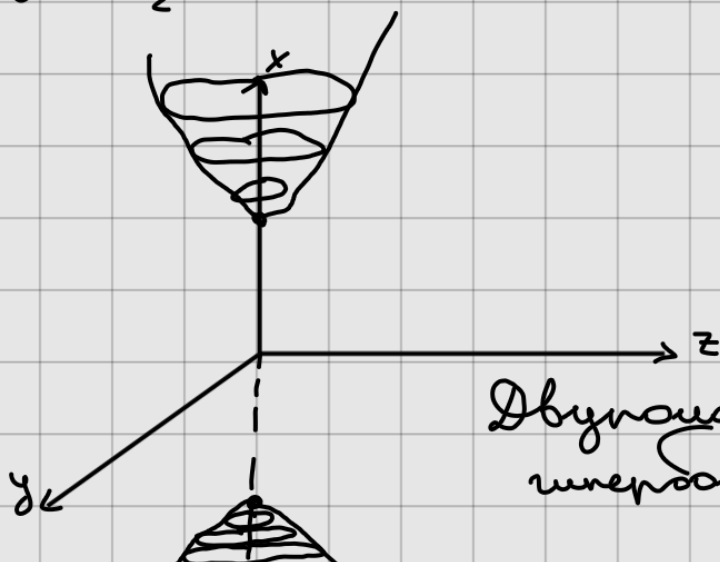
2.3.

$$4y^2 + 3z^2 = 6x^2 - 2 = 0$$

вдоль оси Ox

$$4y^2 + 3z^2 - 6x^2 = -2 \quad | : (-2)$$

$$3x^2 - 2y^2 - \frac{3}{2}z^2 = 1$$



$$\frac{x^2}{a^2} - \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$$

Двухсторонний
гиперболоид
вдоль оси Ox

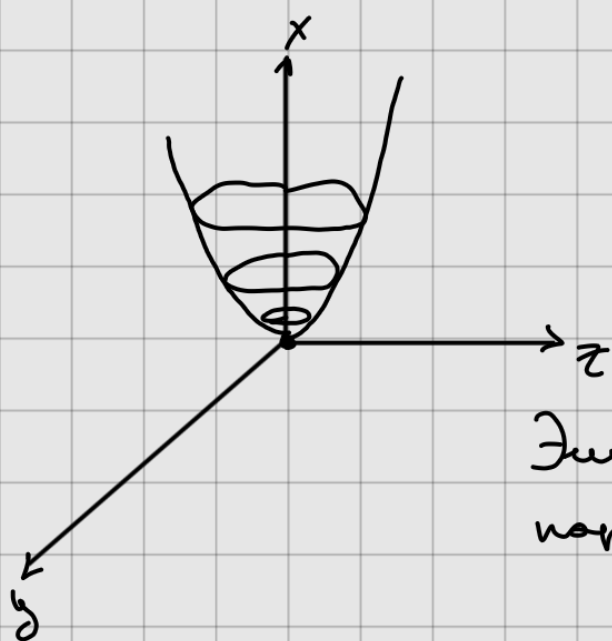
2.4. $3z^2 + y^2 - 4x = 0$

$$3z^2 + y^2 = 4x$$

$$\frac{3}{4}z^2 + \frac{y^2}{4} = x$$

$$\frac{y^2}{b^2} + \frac{z^2}{c^2} = x$$

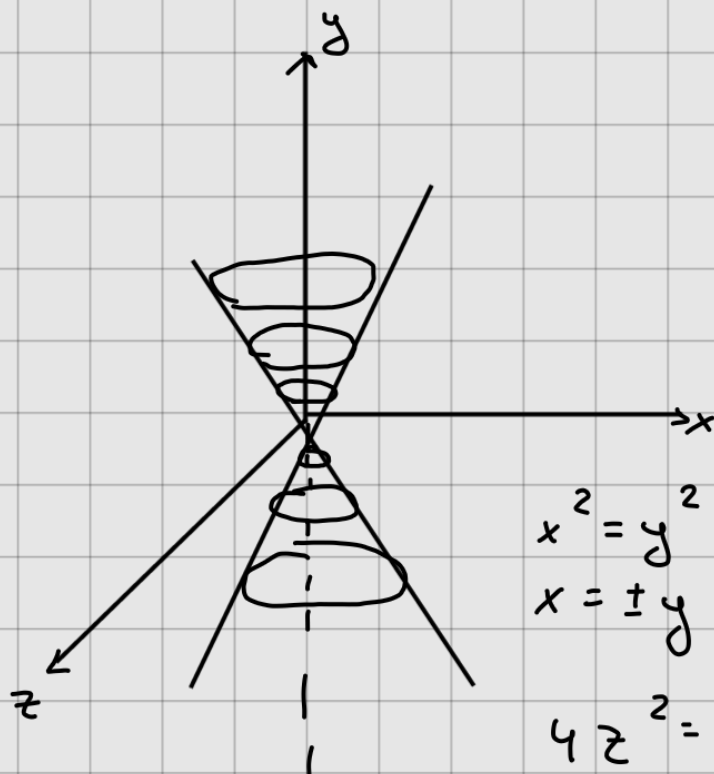
$$b=2, c=\sqrt{\frac{4}{3}}$$



Эллиптический
параболоид

2.5. $x^2 + 4z^2 - y^2 = 0$

$$x^2 + 4z^2 = y^2$$



$$x^2 = y^2$$

$$x = \pm y$$

$$4z^2 = y^2$$

$$2z = \pm y$$

конус

3. Излишние поверхности в трёх-
мерном пространстве.

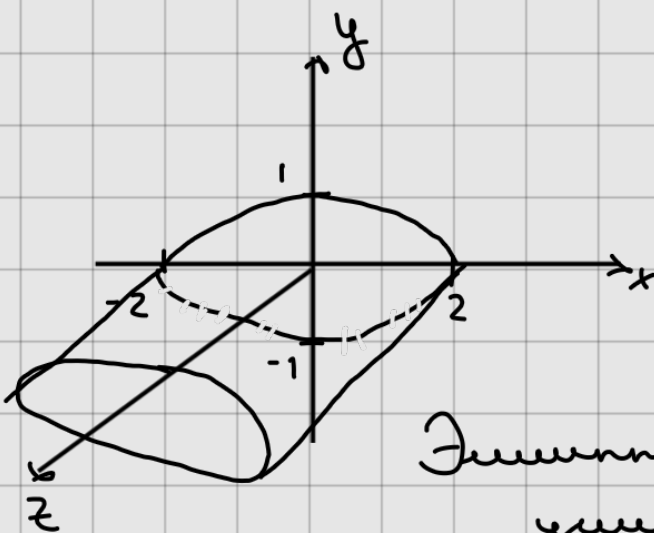
3.1. $x^2 + 4y^2 - 4 = 0$

$$x^2 + 4y^2 = 4$$

$$\frac{x^2}{4} + y^2 = 1$$

$$a=2, b=1$$

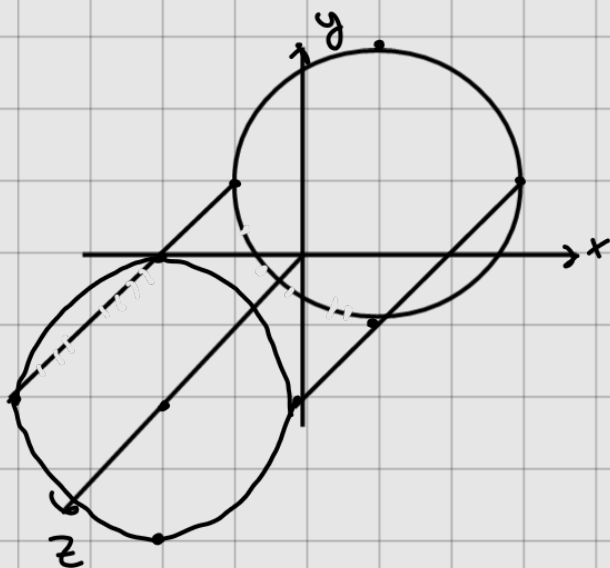
Эллипс в xOy



Эллиптический
цилиндр.

3.2. $x^2 - 2x + y^2 - 2y - 2 = 0$

$$(x-1)^2 + (y-1)^2 = 4$$

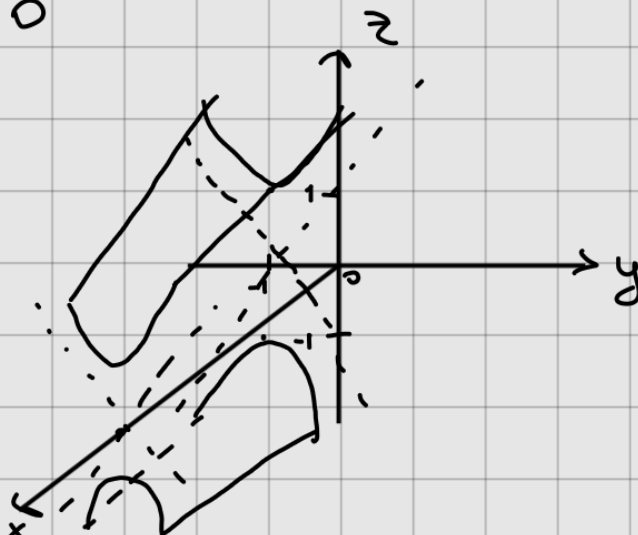


3.3. $z^2 - y^2 - 2y - 2 = 0$

$$z^2 - y^2 - 2y - 1 - 1 = 0$$

$$z^2 - (y+1)^2 = 1$$

$$\frac{z^2}{a^2} - \frac{(y+1)^2}{b^2} = 1$$



ф.л.и (программирование)

16.02.26

1. Построить линии (поверхности) уровня.

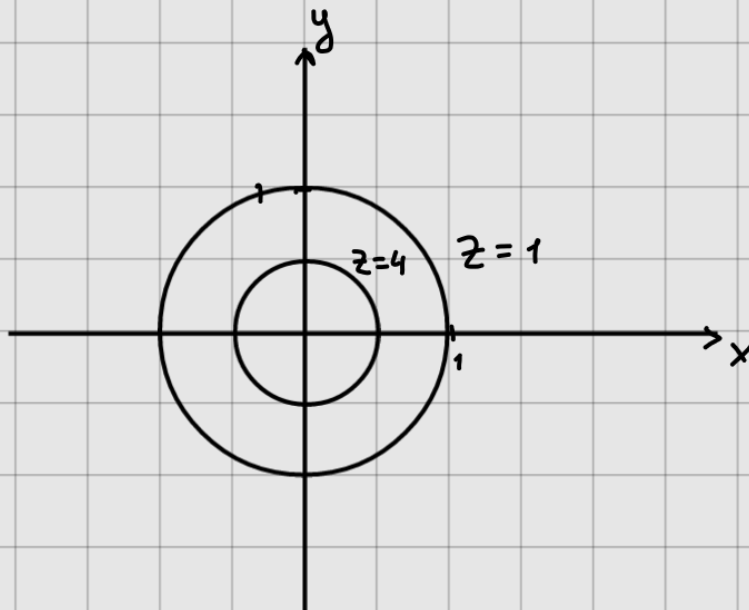
1.1. $z = \frac{1}{x^2 + y^2}$

$z = 0: \frac{1}{x^2 + y^2} = 0 \Rightarrow \text{нет решений}$

$z = 1: \frac{1}{x^2 + y^2} = 1 \Rightarrow x^2 + y^2 = 1$

$z = 4: \frac{1}{x^2 + y^2} = 4 \Rightarrow x^2 + y^2 = \frac{1}{4}$

$z = -1: \frac{1}{x^2 + y^2} = -1 \Rightarrow \text{нет решений}$



1.2. $z = \frac{2x - y - 1}{x + y - 2}$

$z = 0: \frac{2x - y - 1}{x + y - 2} = 0 \Rightarrow 2x - y - 1 = 0$
 $y = 2x - 1$
 $y \neq 2 - x$

$$z=1: \frac{2x-y-1}{x+y-2} = 1 \Rightarrow 2x-y-1 = x+y-2$$

$$2y = x+1$$

$$y = \frac{x+1}{2}$$

$$z=2: \frac{2x-y-1}{x+y-2} = 2 \Rightarrow 2x-y-1 = 2x+2y-4$$

$$3y = 3$$

$$y = 1$$

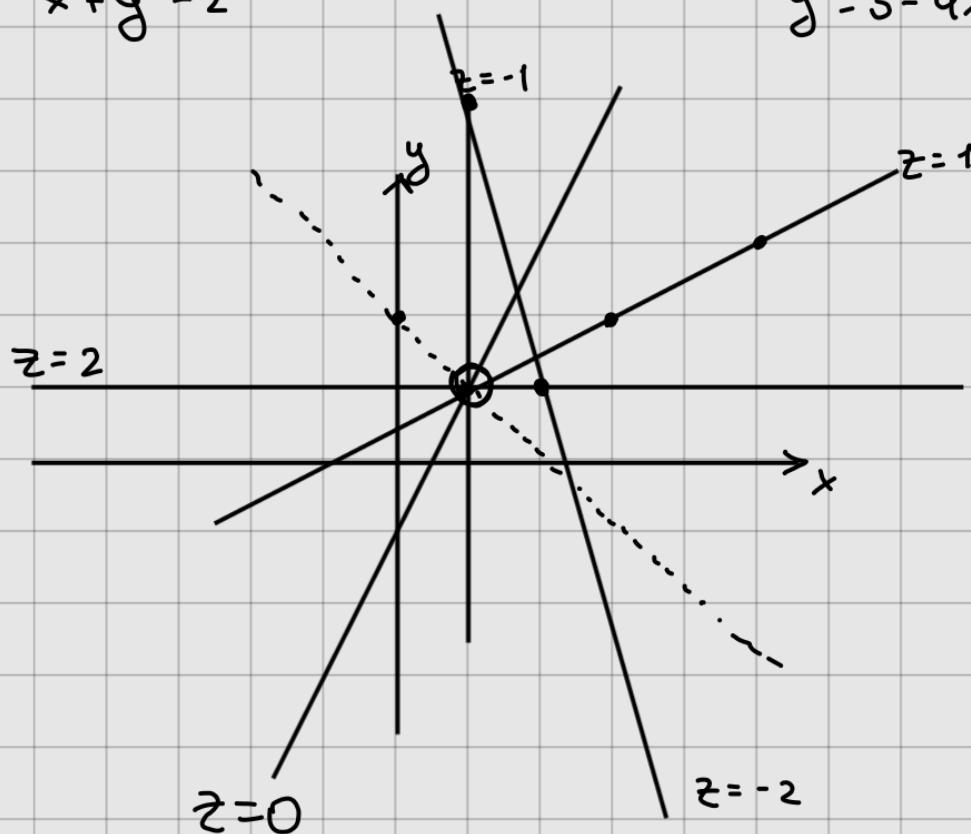
$$z=-1: \frac{2x-y-1}{x+y-2} = -1 \Rightarrow 2x-y-1 = -x-y+2$$

$$3x = 3$$

$$x = 1$$

$$z=-2: \frac{2x-y-1}{x+y-2} = -2 \Rightarrow 2x-y-1 = -2x-2y+4$$

$$y = 5-4x$$



1.3.

$$u = u_n \quad \frac{1 + \sqrt{x^2 + y^2 + z^2}}{1 - \sqrt{x^2 + y^2 + z^2}}$$

$$u = 0 \Rightarrow u_n \quad \frac{1 + \sqrt{x^2 + y^2 + z^2}}{1 - \sqrt{x^2 + y^2 + z^2}} = 0 \Rightarrow \frac{1 + \sqrt{x^2 + y^2 + z^2}}{1 - \sqrt{x^2 + y^2 + z^2}} = 1 \Rightarrow$$

$$\Rightarrow \begin{cases} 1 + \sqrt{x^2 + y^2 + z^2} = 1 - \sqrt{x^2 + y^2 + z^2} \\ 1 - \sqrt{x^2 + y^2 + z^2} \neq 0 \end{cases} \Rightarrow \begin{cases} \sqrt{x^2 + y^2 + z^2} = 0 \\ \sqrt{x^2 + y^2 + z^2} \neq 1 \end{cases} \Rightarrow$$

$$\Rightarrow \begin{cases} x^2 + y^2 + z^2 = 0 \\ x^2 + y^2 + z^2 \neq 1 \end{cases} \Rightarrow x = y = z = 0$$

$$u = u_2 \Rightarrow u_n \quad \frac{1 + \sqrt{x^2 + y^2 + z^2}}{1 - \sqrt{x^2 + y^2 + z^2}} = u_2 \Rightarrow \frac{1 + \sqrt{x^2 + y^2 + z^2}}{1 - \sqrt{x^2 + y^2 + z^2}} = 2,$$

$$1 + \sqrt{x^2 + y^2 + z^2} = 2 - 2\sqrt{x^2 + y^2 + z^2}$$

$$\sqrt{x^2 + y^2 + z^2} = \frac{1}{3} \Rightarrow x^2 + y^2 + z^2 = \frac{1}{9}$$

$$u = u_4 \Rightarrow u_n \quad \frac{1 + \sqrt{x^2 + y^2 + z^2}}{1 - \sqrt{x^2 + y^2 + z^2}} = u_4; \quad \frac{1 + \sqrt{x^2 + y^2 + z^2}}{1 - \sqrt{x^2 + y^2 + z^2}} = 4;$$

$$1 + \sqrt{x^2 + y^2 + z^2} = 4 - 4\sqrt{x^2 + y^2 + z^2}$$

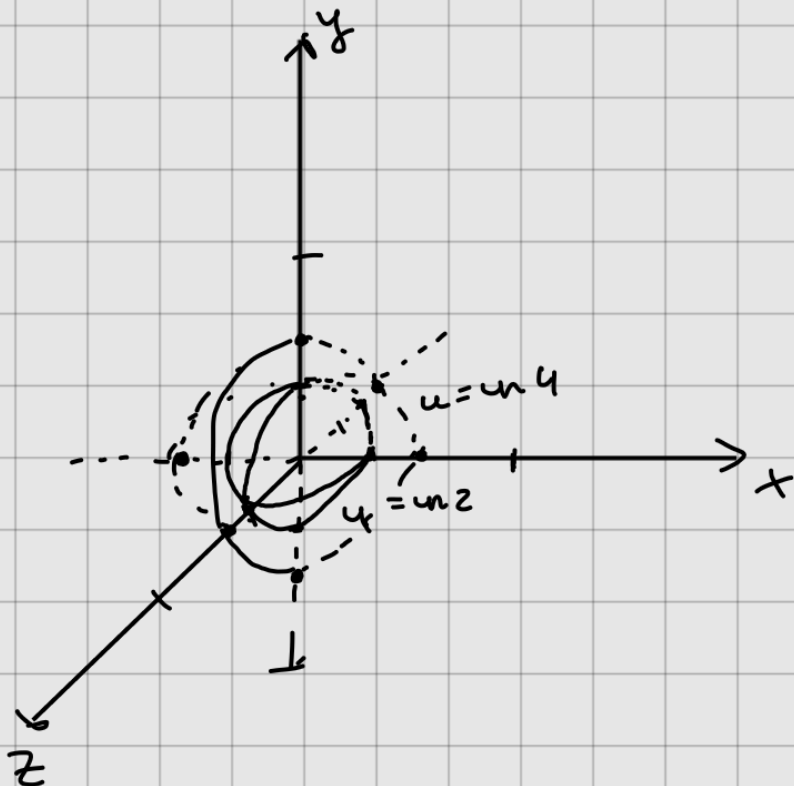
$$5\sqrt{x^2 + y^2 + z^2} = 3; \quad \sqrt{x^2 + y^2 + z^2} = \frac{3}{5}$$

$$x^2 + y^2 + z^2 = \frac{9}{25}$$

$$u = \ln \frac{1}{2} \Rightarrow \ln \frac{1 + \sqrt{x^2 + y^2 + z^2}}{1 - \sqrt{x^2 + y^2 + z^2}} = \ln \frac{1}{2} \Rightarrow$$

$$2 + 2\sqrt{x^2 + y^2 + z^2} = 1 - \sqrt{x^2 + y^2 + z^2} \Rightarrow 3\sqrt{x^2 + y^2 + z^2} = -1$$

$$\sqrt{x^2 + y^2 + z^2} = -\frac{1}{3} \Rightarrow \text{нет решений.}$$



2. Вычислить частные производные 1-го порядка.

2.1. $z(x, y) = (5x^2 \cdot y - y^3 + e^x - 1)^4$

$$\frac{\partial z}{\partial x} = z'_x = 4(5x^2 \cdot y - y^3 + e^x - 1)^3 \cdot (10xy - 0 + e^x)$$

$$\frac{\partial z}{\partial y} = z'_y = 4(5x^2 \cdot y - y^3 + e^x - 1)^3 \cdot (5x^2 - 3y^2)$$

2.2. $z = \frac{x^3 + y^3}{x^2 + y^2}$

$$\frac{\partial z}{\partial x} = z'_x = \frac{3x^2(x^2 + y^2) - (x^3 + y^3) \cdot 2x}{(x^2 + y^2)^2}$$

$$\frac{\partial z}{\partial y} = z'_y = \frac{3y^2(x^2 + y^2) - (x^3 + y^3) \cdot 2y}{(x^2 + y^2)^2}$$

2.3.

$$z = \frac{u}{v} + \frac{v^2}{u^2} = \frac{1}{v} \cdot u + u^{-2} \cdot v^2$$

$$\frac{\partial z}{\partial u} = z'_u = \frac{1}{v} - \frac{2v^2}{u^3}$$

$$\frac{\partial z}{\partial v} = z'_v = -\frac{u}{v^2} + \frac{2v}{u^2}$$

2.4. $z = \arctan \frac{x^2 - y}{2}$

$$z'_x = \frac{1}{1 + \left(\frac{x^2 - y}{2}\right)^2} \cdot \frac{1}{2} \cdot 2x = \frac{x}{1 + \left(\frac{x^2 - y}{2}\right)^2}$$

$$z'_y = \frac{1}{1 + \left(\frac{x^2 - y}{2}\right)^2} \cdot \frac{1}{2} \cdot (-1) = \frac{-1}{2 + 2\left(\frac{x^2 - y}{2}\right)^2}$$

$$2.5. \quad z = e^{-\frac{x}{y}} = e^{-\frac{1}{y}x} = e^{-x \cdot y^{-1}}$$

$$z'_x = e^{-\frac{x}{y}} \cdot \left(-\frac{1}{y}\right)$$

$$z'_y = e^{-\frac{x}{y}} \cdot (-x) \cdot \left(-\frac{1}{y^2}\right) = e^{-\frac{x}{y}} \cdot \frac{x}{y^2}$$

$$2.6. \quad z = \ln(x + \ln y)$$

$$\frac{\partial z}{\partial x} = z'_x = \frac{1}{x + \ln y}$$

$$\frac{\partial z}{\partial y} = z'_y = \frac{1}{x + \ln y} \cdot \frac{1}{y} = \frac{1}{y(x + \ln y)}$$

3. Вычислить частные производные 2-го порядка.

$$3.1. \quad z = x^y$$

Вычисляем ч.п. 1-го порядка:

$$z'_x = y \cdot x^{y-1}$$

$$z'_y = x^y \cdot \ln x$$

Вычисляем ч.п. 2-го порядка:

$$\frac{\partial^2 z}{\partial x^2} = z''_{xx} = y \cdot (y-1) \cdot x^{y-2}$$

$$\frac{\partial^2 z}{\partial y^2} = z''_{yy} = x^y \cdot \ln x \cdot \ln x = x^y \cdot \ln^2 x$$

Смешанные производные:

$$z''_{xy} = (y \cdot x^{y-1})'_y = x^{y-1} + y \cdot x^{y-1} \cdot \ln x$$

$$z''_{yx} = (x^y \cdot \ln x)'_x = y \cdot x^{y-1} \cdot \ln x + x^y \cdot \frac{1}{x} =$$

$$= y \cdot x^{y-1} \cdot \ln x + x^{y-1}$$

Для производной 3-го порядка всего будет

4 различные производные. По формуле z^3 но

4 из них будут повторяться.

3.2.

$$z = e^x (\cos y + 2 \sin y)$$

Возьмем ч.п. 1-го порядка:

$$z'_x = e^x (\cos y + 2 \sin y)$$

$$z'_y = e^x (-\sin y + 2 \cos y)$$

Возьмем ч.п. 2-го порядка:

$$z''_{xx} = e^x (\cos y + 2 \sin y)$$

$$z''_{yy} = e^x (-\cos y - 2\sin y)$$

Смешанное производное:

$$z''_{xy} = (e^x (\cos y + 2\sin y))'_y = e^x (-\sin y + 2\cos y)$$

$$z''_{yx} = (e^x (-\sin y + 2\cos y))'_x = e^x (-\sin y + 2\cos y)$$

3.3. $z = \operatorname{arctg} \frac{y}{x} = \operatorname{arctg}(y \cdot x^{-1})$

Ч. н. первого порядка:

$$z'_x = \frac{1}{1 + \frac{y^2}{x^2}} \cdot \left(-\frac{y}{x^2}\right) = -\frac{y}{x^2 + y^2}$$

$$z'_y = \frac{1}{1 + \frac{y^2}{x^2}} \cdot \frac{1}{x} = \frac{1}{\frac{x^2 + y^2}{x^2}} \cdot \frac{1}{x} =$$

$$= \frac{x}{x^2 + y^2}$$

Ч. н. 2-го порядка:

$$z''_{xx} = \frac{-0 \cdot (x^2 + y^2) + 2y \cdot x}{(x^2 + y^2)^2} = \frac{2yx}{(x^2 + y^2)^2}$$

$$z''_{yy} = \frac{0 \cdot (x^2 + y^2) - 2xy}{(x^2 + y^2)^2} = \frac{-2xy}{(x^2 + y^2)^2}$$

Смешанное производное:

$$z''_{xy} = \left(-\frac{y}{x^2+y^2} \right)'_y = \frac{-1(x^2+y^2) + 2y^2}{(x^2+y^2)^2} = \frac{y^2-x^2}{(x^2+y^2)^2}$$

$$z''_{yx} = \left(\frac{x}{x^2+y^2} \right)'_x = \frac{x^2+y^2 - 2x^2}{(x^2+y^2)^2} = \frac{y^2-x^2}{(x^2+y^2)^2}$$

Дифференцирование, касательные и нормали, производные функции нескольких переменных.

1. Дифференцирование 1-го переменных функций.

$$u = u(x_1, x_2, \dots, x_n)$$

$$du = \frac{\partial u}{\partial x_1} \cdot dx_1 + \frac{\partial u}{\partial x_2} \cdot dx_2 + \dots + \frac{\partial u}{\partial x_n} \cdot dx_n$$

1.1. $u = u(x^4 + 2y^2 - z)$ $du = ?$

$$\frac{\partial u}{\partial x} = \frac{4x^3}{x^4 + 2y^2 - z}$$

$$\frac{\partial u}{\partial y} = \frac{4y}{x^4 + 2y^2 - z}$$

$$\frac{\partial u}{\partial z} = \frac{-1}{x^4 + 2y^2 - z}$$

$$du = \frac{4x^3}{x^4 + 2y^2 - z} dx + \frac{4y}{x^4 + 2y^2 - z} dy + \frac{-1}{x^4 + 2y^2 - z} dz$$

1.2. $z = \frac{x+y}{x-y}$ $dz = ?$

$$\frac{\partial z}{\partial x} = \frac{x-y - x-y}{(x-y)^2} = \frac{-2y}{(x-y)^2}$$

$$\frac{\partial z}{\partial y} = \frac{x-y + x+y}{(x-y)^2} = \frac{2x}{(x-y)^2}$$

$$\begin{aligned} dz &= \frac{\partial z}{\partial x} \cdot dx + \frac{\partial z}{\partial y} \cdot dy = \frac{-2y dx}{(x-y)^2} + \frac{2x dy}{(x-y)^2} = \\ &= \frac{2x dy - 2y dx}{(x-y)^2} \end{aligned}$$

1.3. $z = (x-2)^2 \cos(4y)$ $dz = ?$

$$\frac{\partial z}{\partial x} = 2(x-2) \cdot \cos(4y)$$

$$\frac{\partial z}{\partial y} = (x-2)^2 \cdot (-\sin(4y)) \cdot 4 = -4(x-2)^2 \cdot \sin(4y)$$

$$dz = \frac{\partial z}{\partial x} \cdot dx + \frac{\partial z}{\partial y} \cdot dy = (2x-4) \cos 4y \cdot dx - 4(x-2)^2 \sin 4y \cdot dy$$

$$x \sin(4y) dy$$

1.4. $z = e^{xy}$, $x=1, y=1, \Delta x=0,15, \Delta y=0,1$

$$\frac{\partial z}{\partial x} = y \cdot e^{xy}$$

$$\frac{\partial z}{\partial y} = x \cdot e^{xy}$$

$$dz = \frac{\partial z}{\partial x} \cdot dx + \frac{\partial z}{\partial y} \cdot dy = y \cdot e^{xy} \cdot dx + x \cdot e^{xy} \cdot dy$$

$$+ y \cdot e^{xy} \cdot dy = 1 \cdot e^1 \cdot 0,15 + 1 \cdot e^{1 \cdot 1} \cdot 0,1 =$$

$$= 0,15e + 0,1e = 0,25e$$

2. Косангендроне максималъ и нормалъ
гуре функциямъ двух переменныхъ.

$$\text{нов-мъ } F(x, y, z) = 0 \quad M_0(x_0, y_0, z_0) \in F$$

$$\text{кас. ме-мъ : } F'_x(M_0)(x-x_0) + F'_y(M_0)(y-y_0) + \\ + F'_z(M_0)(z-z_0) = 0$$

$$\text{формула: } \frac{x-x_0}{F'_x(M_0)} = \frac{y-y_0}{F'_y(M_0)} = \frac{z-z_0}{F'_z(M_0)}$$

$$2.1. \quad z = 4x^2 - 3y^2 \quad M_0(1, 1, 1)$$

$$z - 4x^2 + 3y^2 = 0$$

$$\frac{\partial z}{\partial x} = -8x$$

$$\frac{\partial z}{\partial y} = 6y$$

$$\frac{\partial z}{\partial z} = 1$$

$$F'_x(M_0) = -8$$

$$F'_y(M_0) = 6$$

$$F'_z(M_0) = 1$$

$$K: -8(x-1) + 6(y-1) + 1(z-1) = 0$$

$$-8x + 8 + 6y - 6 + z - 1 = 0$$

$$-8x + 6y + z + 1 = 0$$

$$N: \frac{x-1}{-8} = \frac{y-1}{6} = \frac{z-1}{1}$$

$$2.2. \quad 2x^2 + 4z^2 = y^2 \quad M_0(0; -2; 1)$$

$$2x^2 + 4z^2 - y^2 = 0$$

$$F'_x = 4x$$

$$F'_y = -2y$$

$$F'_z = 8z$$

$$F'_x(M_0) = 0$$

$$F'_y(M_0) = 4$$

$$F'_z(M_0) = 8$$

$$K: 0 \cdot (x-0) + 4(y+2) + 8(z-1)$$

$$N: \frac{x-0}{0} = \frac{y+2}{4} = \frac{z-1}{8}$$

3. Показать, что $\mathcal{F} \mathcal{F} \mathcal{T}$ удовлетворяет уравнению.

$$3.1. \quad z = \frac{y}{y^2 - x^2}$$

$$\frac{\partial^2 z}{\partial x^2} = \frac{\partial^2 z}{\partial y^2}$$

$$\frac{\partial z}{\partial x} = \frac{2xy}{(y^2 - x^2)^2} \quad \frac{\partial^2 z}{\partial x^2} = \frac{2y(y^2 - x^2)^2 - 2xy(-4x(y^2 - x^2))}{(y^2 - x^2)^4}$$

$$= \frac{2y(y^2 - x^2)^2 + 8xy(y^2 - x^2)}{(y^2 - x^2)^4} = \frac{2y(y^2 - x^2) + 8xy}{(y^2 - x^2)^3}$$

$$= \frac{2y^3 - 2x^2y + 8x^2y}{(y^2 - x^2)^3} = \frac{2y^3 + 6x^2y}{(y^2 - x^2)^3}$$

$$\frac{\partial z}{\partial y} = \frac{y^2 - x^2 - y \cdot 2y}{(y^2 - x^2)^2} = \frac{-(y^2 + x^2)}{(y^2 - x^2)^2}$$

$$\frac{\partial^2 z}{\partial y^2} = \frac{-2y(y^2 - x^2)^2 + (y^2 + x^2) \cdot 2(y^2 - x^2) \cdot 2y}{(y^2 - x^2)^4} =$$

$$= \frac{-2y(y^2 - x^2) + 4y(y^2 + x^2)}{(y^2 - x^2)^3} = \frac{-2y^3 + 2x^2y + 4y^3 + 4x^2y}{(y^2 - x^2)^3}$$

$$= \frac{2y^3 + 6x^2y}{(y^2 - x^2)^3}$$

$$\Rightarrow \frac{\partial^2 z}{\partial x^2} = \frac{\partial^2 z}{\partial y^2} \quad \blacksquare$$

$$3.2. \quad u = \frac{1}{\sqrt{x^2 + y^2 + z^2}} \quad \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = 0$$

$$\frac{\partial u}{\partial x} = \left((x^2 + y^2 + z^2)^{-\frac{1}{2}} \right)'_x = -\frac{1}{2} (x^2 + y^2 + z^2)^{-\frac{3}{2}} \cdot 2x =$$

$$= -x(x^2 + y^2 + z^2)^{-\frac{3}{2}}$$

$$\frac{\partial^2 u}{\partial x^2} = -(x^2 + y^2 + z^2)^{-\frac{3}{2}} + \frac{3}{2} x(x^2 + y^2 + z^2)^{-\frac{5}{2}} \cdot 2x =$$

$$= -(x^2 + y^2 + z^2)^{-\frac{3}{2}} + 3x^2(x^2 + y^2 + z^2)^{-\frac{5}{2}}$$

В силу симметрии $\frac{\partial^2 u}{\partial y^2}$ и $\frac{\partial^2 u}{\partial z^2}$ будут так $\frac{\partial^2 u}{\partial x^2}$,

только с другой переменной.

$$\frac{\partial^2 u}{\partial y^2} = -(x^2 + y^2 + z^2)^{-\frac{3}{2}} + 3y^2(x^2 + y^2 + z^2)^{-\frac{5}{2}}$$

$$\frac{\partial^2 u}{\partial z^2} = -(x^2 + y^2 + z^2)^{-\frac{3}{2}} + 3z^2(x^2 + y^2 + z^2)^{-\frac{5}{2}}$$

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = \frac{-3}{\sqrt{(x^2+y^2+z^2)^3}} + \frac{3(x^2+y^2+z^2)}{\sqrt{(x^2+y^2+z^2)^5}} =$$

$$= \frac{-3}{\sqrt{(x^2+y^2+z^2)^3}} + \frac{3}{\sqrt{(x^2+y^2+z^2)^3}} = 0 \Rightarrow$$

$$\Rightarrow \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = 0 \quad \blacksquare$$

3.3. $z = e^x (x \cos y - y \sin y)$

$$\frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = 0$$

$$\frac{\partial z}{\partial x} = e^x (x \cos y - y \sin y) + e^x (\cos y)$$

$$\frac{\partial^2 z}{\partial x^2} = e^x (x \cos y - y \sin y) + e^x (\cos y) + \cos y \cdot e^x =$$

$$= e^x (x \cos y - y \sin y) + 2e^x \cdot \cos y = e^x \cdot x \cos y - e^x y \sin y + 2e^x \cos y$$

$$\frac{\partial z}{\partial y} = -e^x (-x \sin y - \sin y - y \cos y) =$$

$$= e^x (x \sin y + \sin y + y \cos y)$$

$$\frac{\partial^2 z}{\partial y^2} = e^x (x \cos y + \cos y + \cos y - y \sin y) = e^x (x \cos y + 2 \cos y - y \sin y) = e^x \cdot x \cos y - e^x y \sin y + 2e^x \cos y \quad \blacksquare$$

4. Частные производные функции второго порядка.

4.1. $z = \ln(x^2 + y^2)$ $\frac{\partial^3 z}{\partial x \partial y^2}$

$$\frac{\partial z}{\partial x} = \frac{2x}{x^2 + y^2} \quad \frac{\partial^2 z}{\partial x \partial y} = \frac{-2x \cdot 2y}{(x^2 + y^2)^2} = \frac{-4xy}{(x^2 + y^2)^2}$$

$$\frac{\partial^3 z}{\partial x \partial y^2} = \frac{-4x(x^2 + y^2)^2 + 4xy \cdot 2(x^2 + y^2) \cdot 2y}{(x^2 + y^2)^4} =$$

$$= \frac{-4x(x^2 + y^2) + 16xy^2}{(x^2 + y^2)^3} = \frac{-4x^3 - 4xy^2 + 16xy^2}{(x^2 + y^2)^3} =$$

$$= \frac{12xy^2 - 4x^3}{(x^2 + y^2)^3}$$

4.2. $z = e^{xy^2}$ $\frac{\partial^3 z}{\partial x^2 \partial y} = ?$

$$\frac{\partial z}{\partial x} = e^{xy^2} \cdot y^2$$

$$\frac{\partial^2 z}{\partial x^2} = y^2 \cdot e^{xy^2} \cdot y^2 = y^4 \cdot e^{xy^2}$$

$$\begin{aligned} \frac{\partial^3 z}{\partial x^2 \partial y} &= 4y^3 \cdot e^{xy^2} + y^4 \cdot e^{xy^2} \cdot 2y = \\ &= 4y^3 e^{xy^2} + 2y^5 e^{xy^2} \end{aligned}$$

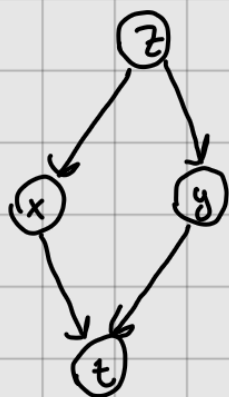
Сложные ф-ые, простые ф-ые, проз по композициям
и прозект, ф-а Тейлора

1. Дифференцирование сложной ф-е.

1.1. $z = e^{2x-y}$

$$\frac{dz}{dt} = ?$$

$$x = \sin(t), \quad y = t^2$$



$$\frac{\partial z}{\partial x} = 2e^{2x-y}$$

$$\frac{\partial z}{\partial y} = -e^{2x-y}$$

$$\frac{dx}{dt} = \cos t$$

$$\frac{dy}{dt} = 2t$$

$$\begin{aligned} \frac{dz}{dt} &= \frac{\partial z}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial z}{\partial y} \cdot \frac{dy}{dt} = 2e^{2x-y} \cos t - \\ &- e^{2x-y} \cdot 2t = 2e^{2x-y} (\cos t - t) \end{aligned}$$

1.2. $u = \ln(x-y), \quad x = 3t, \quad y = 2t^3$ $\frac{du}{dt} = ?$



$$\frac{\partial u}{\partial x} = \frac{1}{x-y}$$

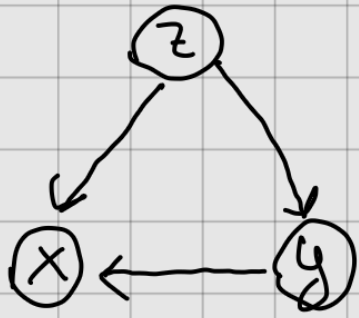
$$\frac{\partial u}{\partial y} = -\frac{1}{x-y}$$

$$\frac{dx}{dt} = 3$$

$$\frac{dy}{dt} = 6t$$

$$\begin{aligned} \frac{du}{dt} &= \frac{\partial u}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial u}{\partial y} \cdot \frac{dy}{dt} = \frac{3}{x-y} - \frac{6t^2}{x-y} = \\ &= \frac{3 - 6t^2}{3t - 2t^3} \end{aligned}$$

1.3. $z = \arctan\left(\frac{x}{y}\right), \quad y = \sqrt{x^2+1}, \quad \frac{dz}{dx} = ?$



$$\frac{dz}{dx} = \frac{\partial z}{\partial x} + \frac{\partial z}{\partial y} \cdot \frac{dy}{dx}$$

$$\frac{\partial z}{\partial x} = \frac{1}{1 + \frac{x^2}{y^2}} \cdot \frac{1}{y} = \frac{1}{\frac{y^2+x^2}{y^2}} \cdot \frac{1}{y} =$$

$$= \frac{y}{y^2+x^2}$$

$$\frac{\partial z}{\partial y} = -\frac{1}{1 + \frac{x^2}{y^2}} \cdot \frac{x}{y^2} = -\frac{x}{\frac{y^2+x^2}{y^2} \cdot y^2} = -\frac{x}{x^2+y^2}$$

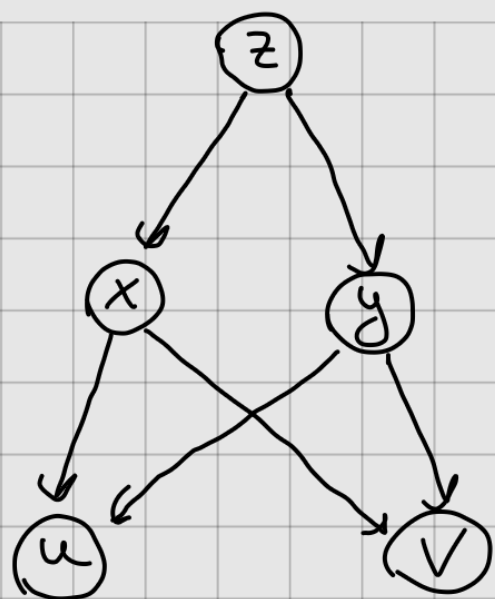
$$\frac{dy}{dx} = \frac{2x}{2\sqrt{x^2+1}} = \frac{x}{\sqrt{x^2+1}}$$

$$\frac{dz}{dx} = \frac{y}{x^2+y^2} - \frac{x}{x^2+y^2} \cdot \frac{x}{\sqrt{x^2+1}} = \frac{1}{x^2+y^2} \left(y - \frac{x^2}{\sqrt{x^2+1}} \right) =$$

$$= \frac{1}{x^2+x^2+1} \cdot \left(\sqrt{x^2+1} - \frac{x^2}{\sqrt{x^2+1}} \right) = \frac{1}{2x^2+1} \left(\frac{x^2+1-x^2}{\sqrt{x^2+1}} \right) =$$

$$= \frac{1}{(2x^2+1)(\sqrt{x^2+1})}$$

1.4 $z = x^2 \cdot y - y^2 \cdot x \quad x = 4 \cdot \cos v \quad y = u \sin v$



$$\frac{\partial z}{\partial u}, \frac{\partial z}{\partial v} = ?$$

$$\frac{\partial z}{\partial u} = \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial u} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial u}$$

$$\frac{\partial z}{\partial v} = \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial v} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial v}$$

$$\frac{\partial z}{\partial x} = 2xy - y^2$$

$$\frac{\partial z}{\partial y} = x^2 - 2yx$$

$$\frac{\partial x}{\partial u} = \cos v$$

$$\frac{\partial y}{\partial u} = \sin v$$

$$\frac{\partial x}{\partial v} = -u \sin v$$

$$\frac{\partial y}{\partial v} = u \cos v$$

$$\frac{\partial z}{\partial u} = (2xy - y^2) \cos v + (x^2 - 2yx) \sin v =$$

$$= (2u \cos v u \sin v - u^2 \sin^2 v) \cos v + (u^2 \cos^2 v - 2u \cos v u \sin v) \sin v =$$

$$2u^2 \cos^2 v \sin v - u^2 \sin^2 v \cos v + u^2 \cos^2 v \sin v - 2u^2 \cos v \sin^2 v =$$

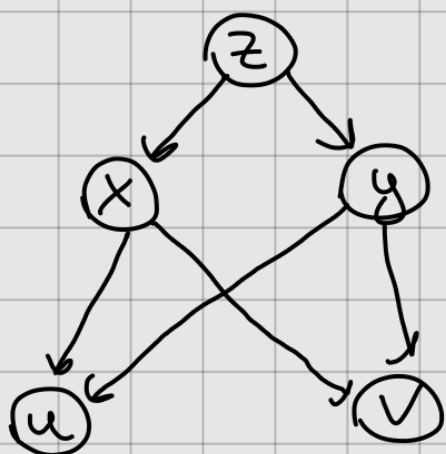
$$3u^2 \cos^2 v \sin v - 3u^2 \sin^2 v \cos v =$$

$$= 3u^2 \cos v \sin v (\cos v - \sin v)$$

$$\frac{\partial z}{\partial v} = -(2xy - y^2) \cdot u \sin v + (x^2 - 2yx) \cdot u \cos v =$$

$$\begin{aligned}
 &= - (2u^2 \cos v \sin v - u^2 \sin^2 v) \cdot u \sin v + (u^2 \cos^2 v - \\
 &- 2u^2 \sin v \cos v) u \cos v = -2u^3 \cos v \sin^2 v + u^3 \sin^3 v + \\
 &+ u^3 \cos^3 v - 2u^3 \cos^2 v \sin v = u^3 (\sin^3 v + \cos^3 v - \\
 &- 2 \sin v \cos v (\sin v + \cos v))
 \end{aligned}$$

1.5. $z = x^2 \cdot \ln(y)$, $x = 3u - 2v$, $y = 2u - 3v$



$$\frac{\partial z}{\partial u}, \frac{\partial z}{\partial v} = ?$$

$$\frac{\partial z}{\partial u} = \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial u} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial u}$$

$$\frac{\partial z}{\partial v} = \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial v} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial v}$$

$$\frac{\partial z}{\partial x} = 2x \ln y \quad \frac{\partial x}{\partial u} = 3 \quad \frac{\partial z}{\partial y} = \frac{x^2}{y} \quad \frac{\partial y}{\partial u} = 2$$

$$\frac{\partial x}{\partial v} = -2 \quad \frac{\partial y}{\partial v} = -3$$

$$\frac{\partial z}{\partial u} = 6x \ln y + \frac{2x^2}{y} = 6(3u - 2v) \cdot \ln(2u - 3v) +$$

$$+ \frac{2(3u - 2v)^2}{2u - 3v} = (18u - 12v) \ln(2u - 3v) + \frac{2(3u - 2v)^2}{2u - 3v}$$

$$\frac{\partial z}{\partial v} = -4x \ln y - \frac{3x^2}{y} = -4(3u-2v) \ln(2y-3v) -$$

$$- \frac{3(3u-2v)^2}{2u-3v} - (8v-12u) \ln(2u-3v) - \frac{3(3u-2v)^2}{2u-3v}$$

2. Дифференцирование неявных ф-ций.

2.1. $x^2 \cos y + y^3 \cos z + z^4 \cos x = 2$ $x^2 \cos y + y^3 \cos z + z^4 \cos x = 0$
 $z'_x, z'_y = ?$

$$z'_x = \frac{F'_x}{-F'_z}$$

$$z'_y = \frac{F'_y}{-F'_z}$$

$$F'_x = 2x \cos y - z^4 \sin x$$

$$F'_y = -x^2 \sin y + 3y^2 \cos z$$

$$F'_z = -y^3 \sin z + 4z^3 \cos x$$

$$z'_x = \frac{2x \cos y - z^4 \sin x}{y^3 \sin z - 4z^3 \cos x}$$

$$z'_y = \frac{3y^2 \cos z - x^2 \sin y}{y^3 \sin z - 4z^3 \cos x}$$

Дифференцирование.

$$2x \cos y - y^3 \sin z \cdot z'_x + 4z^3 \cdot z'_x \cdot \cos x - z^4 \sin x = 0$$

2.2 $e^{-x^2 y} + \ln(z) + \sqrt{z} = 0$

$$F'_x = -2xy \cdot e^{-x^2y}$$

$$F'_y = -x^2 e^{-x^2y}$$

$$F'_z = \frac{1}{z} + \frac{1}{2\sqrt{z}}$$

$$z'_x = \frac{2xy \cdot e^{-x^2y}}{\frac{1}{z} + \frac{1}{2\sqrt{z}}}$$

$$z'_y = \frac{x^2 e^{-x^2y}}{\frac{1}{z} + \frac{1}{2\sqrt{z}}}$$

2.3. $y^2 + \ln(x - 2z) + z^2 = 0$

$$F'_x = \frac{1}{\cos^2(x - 2z)}$$

$$F'_y = 2y$$

$$F'_z = -\frac{2}{\cos^2(x - 2z)} + 2z$$

$$z'_x = \frac{\frac{1}{\cos^2(x - 2z)}}{\frac{2}{\cos^2(x - 2z)} + 2z} = \frac{1}{2 - 2z \cos^2(x - 2z)}$$

$$z'_y = \frac{2y}{\frac{2}{\cos^2(x - 2z)} + 2z} = \frac{2y \cos^2(x - 2z)}{2 - 2z \cos^2(x - 2z)}$$

3. Производные по направлению и градиент

3.1. $u = \frac{x^2}{y} + \frac{y^2}{z} + \frac{z^2}{x} - xyz$

$$M_0(1;2;1) \quad \frac{\partial u}{\partial L}, \text{ wgl } \vec{L} = \{1;2;2\}$$

$$\text{bestimme } \overrightarrow{\text{grad} u}(M_0) \text{ u } |\overrightarrow{\text{grad} u}|$$

$$\frac{\partial u}{\partial L}(M_0) = \frac{\partial u}{\partial x} \Big|_{M_0} \cdot \cos \alpha + \frac{\partial u}{\partial y} \Big|_{M_0} \cdot \cos \beta + \frac{\partial u}{\partial z} \Big|_{M_0} \cdot \cos \gamma$$

$$\times \cos \gamma$$

$$\overrightarrow{\text{grad} u} = \left\{ \frac{\partial u}{\partial x}; \frac{\partial u}{\partial y}; \frac{\partial u}{\partial z} \right\}$$

$$\frac{\partial u}{\partial x} = \frac{2x}{y} - \frac{z^2}{x^2} - yz \quad \frac{\partial u}{\partial x} \Big|_{M_0} = \frac{2 \cdot 1}{2} - \frac{1}{1} - 2 =$$

$$= -2$$

$$\frac{\partial u}{\partial y} = -\frac{x^2}{y^2} + \frac{2y}{z} - xz$$

$$\frac{\partial u}{\partial y} \Big|_{M_0} = -\frac{1}{4} + \frac{4}{1} - 1 \cdot 1 =$$

$$= \frac{11}{4}$$

$$\frac{\partial u}{\partial z} = -\frac{y^2}{z^2} + \frac{2z}{x} - xy$$

$$\frac{\partial u}{\partial z} \Big|_{M_0} = -\frac{4}{1} + \frac{2}{1} - 2 =$$

$$= -4$$

$$|\vec{L}| = \sqrt{1+4+4} = 3$$

$$\vec{e}_L = \left\{ \frac{1}{3}; \frac{2}{3}; \frac{2}{3} \right\}$$

cos α cos β cos γ

$$\frac{\partial u}{\partial L}(M_0) = -\frac{2}{3} + \frac{11}{6} - \frac{8}{3} = -\frac{3}{2}$$

$$\overrightarrow{\text{grad } u} = \left\{ -2; \frac{11}{4}; -4 \right\}$$

$$|\overrightarrow{\text{grad } u}| = \sqrt{4 + \frac{121}{16} + 16} = \sqrt{\frac{441}{4}} = \frac{21}{2}$$

3.2 $z = x^3 - 2x^2y + xy^2 + 1$ $M(1,2), N(4,6)$

$$\vec{L} = \overrightarrow{MN} \quad \frac{\partial z}{\partial L} = \frac{\partial z}{\partial x} \Big|_M \cdot \cos \alpha + \frac{\partial z}{\partial y} \Big|_M \cdot \cos \beta$$

$$\vec{L} = \{ 3; 4 \}$$

$$\frac{\partial z}{\partial x} = 3x^2 - 4xy + y^2 \quad \frac{\partial z}{\partial x} \Big|_M = 3 - 8 + 4 = -1$$

$$\frac{\partial z}{\partial y} = -2x^2 + 2xy \quad \frac{\partial z}{\partial y} \Big|_M = -2 + 4 = 2$$

$$|\vec{L}| = \sqrt{9 + 16} = \sqrt{25} = 5$$

$$\vec{e}_L = \left\{ \underset{\cos \alpha}{\frac{3}{5}}; \underset{\cos \beta}{\frac{4}{5}} \right\}$$

$$\frac{\partial z}{\partial L} \Big|_M = -1 \cdot \frac{3}{5} + 2 \cdot \frac{4}{5} = 1$$

$$\overrightarrow{\text{grad } z} = \{ -1; 2 \}$$

$$|\overrightarrow{\text{grad } z}| = \sqrt{1 + 4} = \sqrt{5}$$

4. Задача Тейлора

4.1. Разложить функцию $f(x, y) = 2x^2 - 3xy - 4y^3$

в окрестности $M(-2; 1)$ до 2-го порядка

$$f(x, y) \approx f(x_0, y_0) + df(x_0, y_0) + \frac{1}{2!} d^2 f(x_0, y_0)$$

$$f(-2; 1) = 8 + 6 - 4 = 10$$

$$df(-2, 1) = \left. \frac{\partial f}{\partial x} \right|_{M_0} \Delta x + \left. \frac{\partial f}{\partial y} \right|_{M_0} \Delta y =$$

$$= (4x - 3y)|_{M_0} \cdot (x+2) + (-3x - 12y^2)|_{M_0} \cdot (y-1) =$$

$$= -15(x+2) - 6(y-1)$$

$$d^2 f(-2; -1) = \left. \frac{\partial^2 f}{\partial x^2} \right|_{M_0} (\Delta x)^2 + 2 \left. \frac{\partial^2 f}{\partial x \partial y} \right|_{M_0} \Delta x \Delta y + \left. \frac{\partial^2 f}{\partial y^2} \right|_{M_0} (\Delta y)^2$$

$$= 4(x+2)^2 - 6(x+2)(y-1) - 24(y-1)^2$$

$$f(x, y) \approx 10 - 15(x+2) - 6(y-1) + 2(x+2)^2 - 3(x+2)(y-1) - 12(y-1)^2$$

Локальный и абсолютный экстремум ф.з.н (числовой экстремум)

1. Задачи на экстремумы

1.1

$$z = 4(x-y) - x^2 - y^2$$

Находим стационарные точки

$$z'_x = 4 - 2x$$

$$\begin{cases} 4 - 2x = 0 \\ -4 - 2y = 0 \end{cases}$$

$$z'_y = -4 - 2y$$

$$\begin{cases} x = 2 \\ y = -2 \end{cases} \Rightarrow M(2; -2)$$

Проверяем на экстремумы

$$z''_{xx} = -2$$

$$z''_{yy} = -2$$

$$z''_{xy} = 0$$

$$\Delta_2 = \begin{vmatrix} -2 & 0 \\ 0 & -2 \end{vmatrix} = 4 > 0 - \text{т. экстремума}$$

$$\Delta_1 = -2 < 0 \Rightarrow \text{т. максимума}$$

$$z(M) = 16 - 4 - 4 = 8$$

$$M(2; -2)$$

1.2.

$$z = x^2 + xy + 2y^2 + x - y + 1$$

$$z'_x = 2x + y + 1$$

$$z'_y = x + 4y - 1$$

$$\textcircled{1} \begin{cases} 2x + y + 1 = 0 \end{cases}$$

$$\textcircled{2} \begin{cases} x + 4y - 1 = 0 \end{cases}$$

$$\textcircled{2} x = 1 - 4y$$

$$\textcircled{1} 2(1-4y) + y + 1 = 0$$

$$2 - 8y + y + 1 = 0$$

$$-7y + 3 = 0$$

$$y = \frac{3}{7}$$

$$\textcircled{2} x = 1 - 4 \cdot \frac{3}{7} = 1 - \frac{12}{7} =$$

$$= -\frac{5}{7}$$

$$M\left(-\frac{5}{7}; \frac{3}{7}\right)$$

$$z''_{xx} = 2$$

$$z''_{yy} = 4$$

$$z''_{xy} = 1$$

$$\Delta_2 = \begin{vmatrix} 2 & 1 \\ 1 & 4 \end{vmatrix} = 7 > 0$$

$$\Delta_1 = 2 > 0 \Rightarrow \text{точка минимума}$$

$$z(M) = \frac{25}{49} - \frac{15}{49} + \frac{18}{49} - \frac{5}{7} - \frac{3}{7} + 1 = \frac{28}{49} - \frac{56}{49} + \frac{49}{49} =$$

$$= \frac{21}{49} = \frac{3}{7}$$

$$1.3 \quad z = 3xy - x^2y - xy^2$$

$$z'_x = 3y - 2xy - y^2$$

$$z'_y = 3x - x^2 - 2yx$$

$$\begin{cases} 3y - 2xy - y^2 = 0 \\ 3x - x^2 - 2yx = 0 \end{cases}$$

$$\textcircled{1} y(3 - 2x - y) = 0$$

$$\textcircled{2} x(3 - x - 2y) = 0$$

$$\textcircled{1} \begin{cases} y = 0 \\ x(3-x) = 0 \end{cases} \quad \text{или}$$

$$\textcircled{2} \begin{cases} x = 0 \\ y(3-y) = 0 \end{cases}$$

$$\begin{cases} y = 0 \\ x = 0 \\ x = 3 \end{cases}$$

$$\begin{cases} x = 0 \\ y = 0 \\ y = 3 \end{cases}$$

$$\Rightarrow M_1(0,0), M_2(3,0), M_3(0,3)$$

$$\begin{cases} 3-2x-y=0 \\ 3-x-2y=0 \end{cases} \Leftrightarrow \begin{cases} y=3-2x \\ 3-x-2(3-2x)=0 \end{cases} \quad (3)$$

$$(3) \quad 3-x-6+4x=0$$

$$3x-3=0$$

$$x=1$$

$$y=3-2=1 \Rightarrow M_4(1;1)$$

$$z''_{xx} = -2y$$

$$z''_{yy} = -2x$$

$$z''_{xy} = 3-2x-2y$$

$$M_1(0;0): \Delta_2 = \begin{vmatrix} 0 & 3 \\ 3 & 0 \end{vmatrix} = -9 < 0 - \text{нет экстремума}$$

$$M_2(3;0): \Delta_2 = \begin{vmatrix} 0 & -3 \\ -3 & -6 \end{vmatrix} = -9 < 0 - \text{нет экстремума}$$

$$M_3(0;3): \Delta_2 = \begin{vmatrix} -6 & -3 \\ -3 & 0 \end{vmatrix} = -9 < 0 - \text{нет экстремума}$$

$$M_4(1;1): \Delta_2 = \begin{vmatrix} -2 & -1 \\ -1 & -2 \end{vmatrix} = 3 > 0 - \text{точка экстремума}$$

$$\Delta_1: -2 < 0 \Rightarrow \text{т. максимума}$$

$$z(M_4) = 3 - 1 - 1 = 1$$

$$1.4 \quad z = x^3 - 2x^2y + 2y^2 + 1$$

$$z'_x = 3x^2 - 4xy$$

$$z'_y = -2x^2 + 4y$$

$$(1) \begin{cases} 3x^2 - 4xy = 0 \end{cases}$$

$$(2) \begin{cases} -2x^2 + 4y = 0 \end{cases}$$

$$\begin{cases} x(3x-4y)=0 \\ y = \frac{x^2}{2} \end{cases}$$

$$x_1 = 0$$

$$y_1 = 0$$

$$x_2 = \frac{4y}{3}$$

$$y_2 = \frac{16y^2}{9 \cdot 2} = \frac{16y^2}{18}$$

$$x_2 = \frac{4 \cdot \frac{9}{8}}{3} = \frac{\frac{36}{8}}{3} = \frac{36}{24} = \frac{3}{2}$$

$$M_1(0;0)$$

$$18y = 16y^2$$

$$18y - 16y^2 = 0$$

$$y(18 - 16y) = 0$$

$$y = 0 \text{ или } y = \frac{18}{16} = \frac{9}{8}$$

$$M_2\left(\frac{3}{2}; \frac{9}{8}\right)$$

$$z''_{xx} = 6x - 4y \quad z''_{yy} = 4 \quad z''_{xy} = -4x$$

$$z''_{xx}(M_2) = 6 \cdot \frac{3}{2} - 4 \cdot \frac{9}{8} = 9 - \frac{9}{2} = \frac{9}{2}$$

$$z''_{yy}(M_2) = 4$$

$$z''_{xy}(M_2) = -4 \cdot \frac{3}{2} = -6$$

$$\Delta_2(M_2) = \begin{vmatrix} \frac{9}{2} & -6 \\ -6 & 4 \end{vmatrix} = 18 - 36 = -18 \Rightarrow \text{Экстремума нет}$$

$$z''_{xx}(M_1) = 0 \quad z''_{yy}(M_1) = 4 \quad z''_{xy}(M_1) = 0$$

$$\Delta_2(M_1) = \begin{vmatrix} 0 & 0 \\ 0 & 4 \end{vmatrix} = 0 - \text{нужно дальнейшее рассуждение}$$

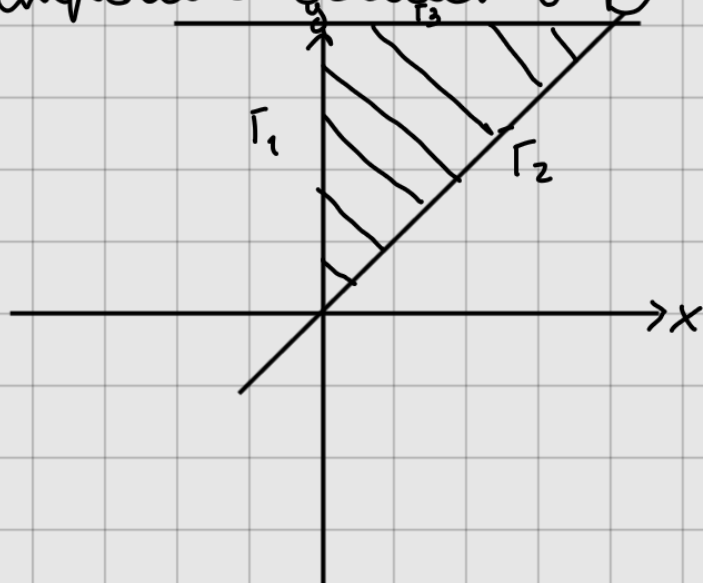
$$y = 0 : z = x^3 + 1 \Rightarrow \text{в т. } x = 0 \text{ экстремума нет.}$$

2. $z \rightarrow \text{absхет}$ (абсолютный экстремум)

2.1. $z = 3x + y - xy$

$D: y = x, y = 4, x = 0$

Построим область D :



$$z'_x = 3 - y \quad z'_y = 1 - x \Rightarrow M(1; 3)$$

$$\Gamma_1: x = 0, y \in [0; 4]$$
$$z = y \Rightarrow M_1(1; 3) \in D$$

$$\Gamma_2: y = x$$
$$z = 3x + x - x^2 = 4x - x^2$$
$$z'_x = 4 - 2x = 0 \Rightarrow x = 2, y = 2 \quad M_2(2; 2) \in D$$

$$\Gamma_3: y = 4, x \in [0; 4]$$

$$z = 3x + 4 - 4x = 4 - x$$

$$z'_x = -1 \neq 0 \Rightarrow \text{нет решений}$$

$$M_3(0; 0), M_4(0; 4), M_5(4; 4)$$

$$z(M_1) = 3+3-3 = 3$$

$$z(M_2) = 6+2-4 = 4$$

$$z(M_3) = 0$$

$$z(M_4) = 4$$

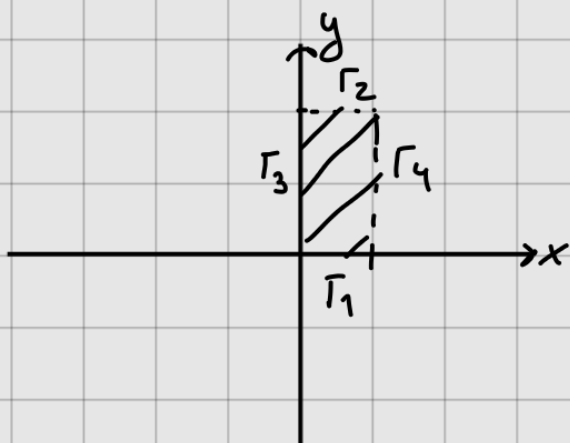
$$z(M_5) = 12+4-16 = 0$$

$$z_{\max} = 4 (M_2; M_4)$$

$$z_{\min} = 0 (M_3; M_5)$$

2.2. $z = x^2 + 2xy - 4x + 8y$

$$D: 0 \leq x \leq 1, 0 \leq y \leq 2$$



$$z'_x = 2x + 2y - 4$$

$$z'_y = 2x + 8$$

$$\begin{cases} 2x + 2y - 4 = 0 \\ 2x + 8 = 0 \end{cases} \Rightarrow \begin{cases} y = 6 \\ x = -4 \end{cases} \Rightarrow$$

$$\Rightarrow M_1(-4; 6) \notin D$$

Исследуем границы:

$$\Gamma_1: y = 0$$

$$z = x^2 - 4x$$

$$z'_x = 2x - 4 = 0$$

$$x = 2, y = 0 \Rightarrow M_2(2; 0) \notin D$$

$$\Gamma_2: y = 2$$

$$z = x^2 + 16$$

$$z'_x = 2x$$

$$\Rightarrow M_3(0; 2) \in D$$

$$\Gamma_3: x = 0$$

$$z = 8y$$

$$z'_y = 8 \neq 0 \Rightarrow \text{нет решений}$$

$$\Gamma_4: x=1$$

$$z = 1 + 2y - 4 + 8y = 10y - 3 \quad z'_y = 10 \neq 0$$

$$M_4(1; 2), \quad M_5(1; 0), \quad M_6(0; 0)$$

$$z(M_3) = 16$$

$$z(M_5) = 1 - 4 = -3$$

$$z(M_4) = 1 + 4 - 4 + 16 = 17$$

$$z(M_6) = 0$$

$$z_{\min} = -3(M_5) \quad z_{\max} = 17(M_4)$$

3. Условный экстремум.

$$3.1. \quad z = 2y^2 + x^2 - 2x + 2; \quad 2x + y + 4 = 0$$

$$\begin{cases} z = 2y^2 + x^2 - 2x + 2 \\ 2x + y + 4 = 0 \end{cases}$$

$$F = 2y^2 + x^2 - 2x + 2 + \lambda(2x + y + 4)$$

$$F'_x = 2x - 2 + 2\lambda$$

$$F'_y = 4y + \lambda$$

$$\begin{cases} 2x - 2 + 2\lambda = 0 \\ 4y + \lambda = 0 \\ 2x + y + 4 = 0 \end{cases} \Leftrightarrow$$

$$x = 1 - \lambda$$

$$y = -\frac{\lambda}{4}$$

$$2(1 - \lambda) - \frac{\lambda}{4} + 4 = 0$$

$$2 - 2\lambda - \frac{\lambda}{4} + 4 = 0$$

$$-\frac{9}{4}\lambda = -6$$

$$\lambda = \frac{24}{9} = \frac{8}{3}$$

$$x_0 = 1 - \frac{8}{3} = -\frac{5}{3}$$

$$y_0 = -\frac{8}{3 \cdot 4} = -\frac{2}{3}$$

$$M\left(-\frac{5}{3}; -\frac{2}{3}\right)$$

$$F''_{xx} = 2$$

$$F''_{xy} = 0$$

$$F''_{yy} = 4$$

$$d^2F(M) = 2dx^2 + 4dy^2 \Rightarrow \text{m. yu. usachunyna}$$

$$F(M) = 2 \cdot \frac{4}{9} + \frac{25}{9} + \frac{10}{3} + 2 = \frac{33}{9} + \frac{30}{9} + 2 =$$

$$= \frac{63}{9} + 2 = 9$$

$$3.2. \quad z = x^2 + y^2 - xy - x - y - 2; \quad x + y + z = 0$$

$$\begin{cases} z = x^2 + y^2 - xy - x - y - 2 \\ x + y + z = 0 \end{cases}$$

$$F = x^2 + y^2 - xy - x - y - 2 + \lambda(x + y + z)$$

$$F'_x = 2x - y - 1 + \lambda$$

$$F'_y = 2y - x - 1 + \lambda$$

$$+ \begin{cases} 2x - y - 1 + \lambda = 0 \\ 2y - x - 1 + \lambda = 0 \\ x + y + z = 0 \end{cases}$$

$$2x + 2y + 2\lambda = 0$$

$$x + y + \lambda = 0$$

$$\Rightarrow \lambda = 2$$

$$x = \frac{y+1-\lambda}{2} = \frac{y-1}{2}$$

$$\frac{y-1}{2} + y + 2 = 0 \Rightarrow y - 1 + 2y + 4 = 0 \Rightarrow 3y = -3 \Rightarrow y = -1$$

$$x = \frac{y-1}{2} = \frac{-2}{2} = -1$$

$$M(-1; -1)$$

$$F''_{xx} = 2 \quad F''_{yy} = 2 \quad F''_{xy} = -1$$

$$d^2F = 2dx^2 - 2dx dy + 2dy^2 = 2dy^2 + 2dy^2 + 2dy^2 = 6dy^2 > 0 \Rightarrow \text{m. maximales} \quad (dx + dy = 0)$$

$$Z(M) = 1 + 1 - 1 + 1 + 1 - 2 = 1$$

$$3.3 \quad Z = x + y$$

$$\frac{1}{x} + \frac{1}{y} = 1$$

$$\begin{cases} Z = x + y \\ \frac{1}{x} + \frac{1}{y} = 1 \end{cases}$$

$$F = x + y + \lambda \left(\frac{1}{x} + \frac{1}{y} - 1 \right) = x + y + \frac{\lambda}{x} +$$

$$+ \frac{\lambda}{y} - \lambda$$

$$F'_x = 1 - \frac{\lambda}{x^2}$$

$$F'_y = 1 - \frac{\lambda}{y^2}$$

$$\begin{cases} 1 - \frac{\lambda}{x^2} = 0 & \frac{\lambda}{x^2} = \frac{\lambda}{y^2} \\ 1 - \frac{\lambda}{y^2} = 0 & y = \pm x \\ \frac{1}{x} + \frac{1}{y} - 1 = 0 \end{cases}$$

1 случай: $x = y$

$$\frac{1}{y} + \frac{1}{y} - 1 = 0 \quad \frac{2}{y} - 1 = 0 \quad y = 2 \Rightarrow x = 2 \Rightarrow M(2; 2)$$

$$\frac{\lambda}{4} = 1 \Rightarrow \lambda = 4$$

2 случай: $x = -y$

$$\frac{1}{x} - \frac{1}{x} - 1 = 0; \quad -1 \neq 0 \quad \text{нет решений}$$

$$F''_{xx} = \frac{2\lambda}{x^3}$$

$$F''_{xy} = 0$$

$$F''_{yy} = \frac{2\lambda}{y^3}$$

$$d^2 F = \frac{2\lambda}{x^3} dx^2 + \frac{2\lambda}{y^3} dy^2$$

$$d^2 F(M) = dx^2 + dy^2 > 0 \Rightarrow M - \text{m. loc. макс.}$$

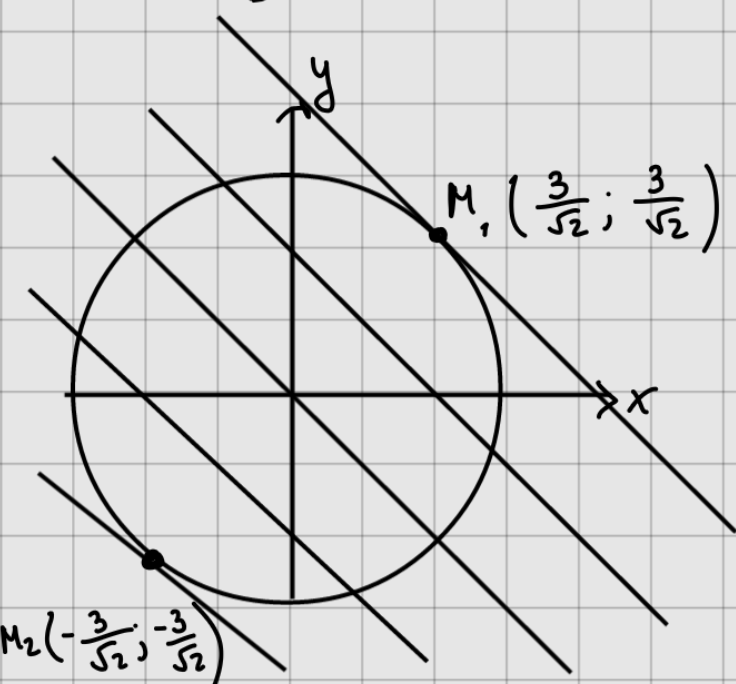
$$z(M) = 2 + 2 = 4$$

Двойные интегралы

1. Оценить двойные интегралы

$$m \cdot S_D \leq \iint_D f(x, y) dS \leq M \cdot S_D$$

1.1. $\iint_D (x+y+2) dS$, $D: x^2 + y^2 \leq 9$



$$z = x + y + 2$$

$$z = 0: y = -x - 2$$

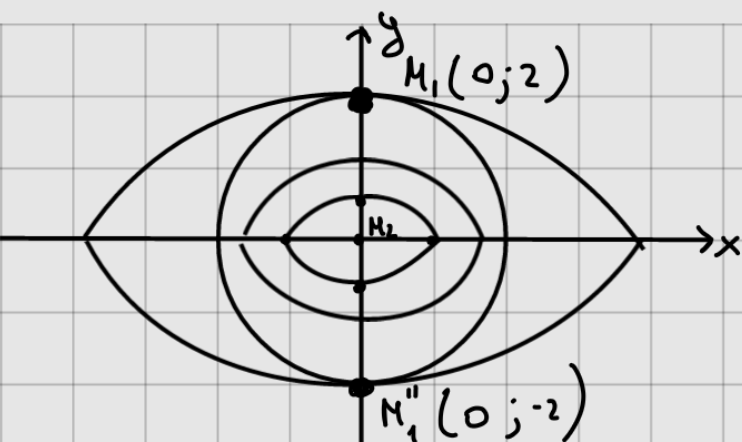
$$z = 2: y = -x$$

$$z(M_1) = \frac{6}{\sqrt{2}} + 2 = 3\sqrt{2} + 2$$

$$z(M_2) = -\frac{6}{\sqrt{2}} + 2 = -3\sqrt{2} + 2$$

$$(-3\sqrt{2} + 2) \cdot 9\pi \leq \iint_D (x+y+2) dS \leq (3\sqrt{2} + 2) \cdot 9\pi$$

1.2. $\iint_D (x^2 + 4y^2 + 5) dS$, $D: x^2 + y^2 \leq 4$



$$z = x^2 + 4y^2 + 5$$

$$z = 5: \quad 4y^2 + x^2 = 0$$

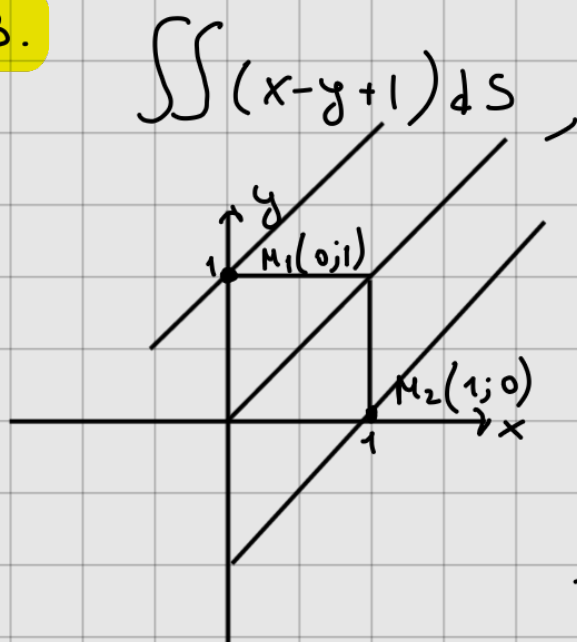
$$z = 6: \quad x^2 + 4y^2 = 1$$

$$z(M_1) = 16 + 5 = 21$$

$$z(M_2) = 5$$

$$5 \cdot 4\pi \leq \iint_D (x^2 + 4y^2 + 5) dS \leq 21 \cdot 4\pi$$

1.3.



$$D: \quad 0 \leq x \leq 1$$

$$0 \leq y \leq 1$$

$$z = x - y + 1$$

$$z = 0: \quad y = x + 1$$

$$z = 1: \quad y = x$$

$$z = 2: \quad y = x - 1$$

$$z(M_1) = 0$$

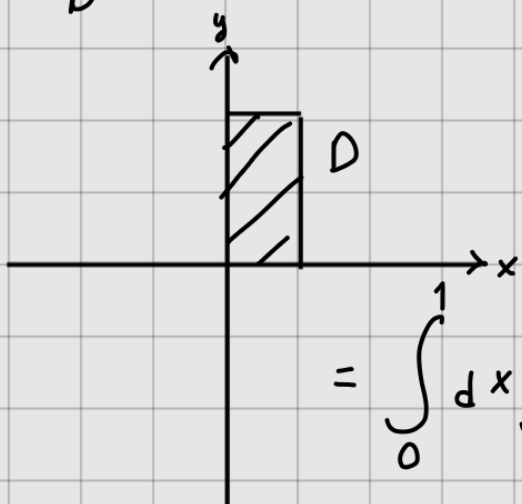
$$z(M_2) = 2$$

$$0 \leq \iint_D (x - y + 1) dS \leq 2 \cdot 1$$

2. Двойные интегралы в криволинейных координатах.

2.1.

$$\iint_D (x+y) dx dy, \quad D: \begin{cases} 0 \leq x \leq 1 \\ 0 \leq y \leq 2 \end{cases}$$



$$\iint_D (x+y) dx dy =$$

$$= \int_0^1 dx \int_0^2 (x+y) dy = \int_0^1 \left(xy + \frac{y^2}{2} \Big|_0^2 \right) dx =$$

$$= \int_0^1 (2x + 2) dx = x^2 + 2x \Big|_0^1 = 1 + 2 = 3$$

2.2.

$$\iint_D e^{x-y} dx dy, \quad D: \begin{cases} 0 \leq x \leq 1 \\ 0 \leq y \leq 1 \end{cases}$$

$$\iint_D e^{x-y} dx dy = \int_0^1 e^x dx \int_0^1 e^{-y} dy = \int_0^1 e^x dx \cdot (-e^{-y} \Big|_0^1)$$

$$= \int_0^1 e^x dx \cdot \left(-\frac{1}{e} + 1 \right) = \left(1 - \frac{1}{e} \right) \cdot e^x \Big|_0^1 = \left(1 - \frac{1}{e} \right) (e - 1)$$

$$= e - 1 - 1 + \frac{1}{e} = e + \frac{1}{e} - 2$$

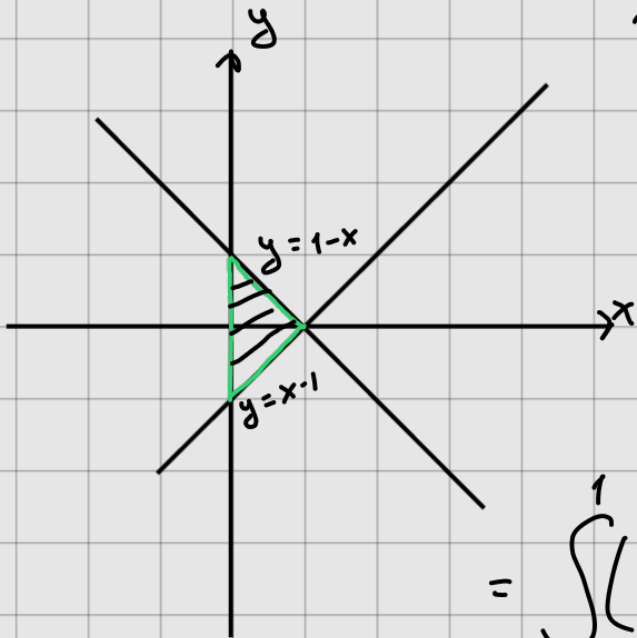
2.3.

$$\iint_D \frac{x^2}{1+y^2} dx dy, \quad D: \begin{cases} 0 \leq x \leq 2 \\ 0 \leq y \leq 1 \end{cases}$$

$$\begin{aligned}
 \iint_D \frac{x^2}{1+y^2} dx dy &= \int_0^1 dy \int_0^2 \frac{x^2}{1+y^2} dx = \int_0^1 \left(\frac{\frac{x^3}{3}}{1+y^2} \Big|_0^2 \right) dy = \\
 &= \int_0^1 \frac{\frac{8}{3}}{1+y^2} dy = \frac{8}{3} \int_0^1 \frac{1}{1+y^2} dy = \frac{8}{3} \operatorname{arctg}(y) \Big|_0^1 = \\
 &= \frac{8}{3} \left(\frac{\pi}{4} - 0 \right) = \frac{8\pi}{12} = \frac{2\pi}{3}
 \end{aligned}$$

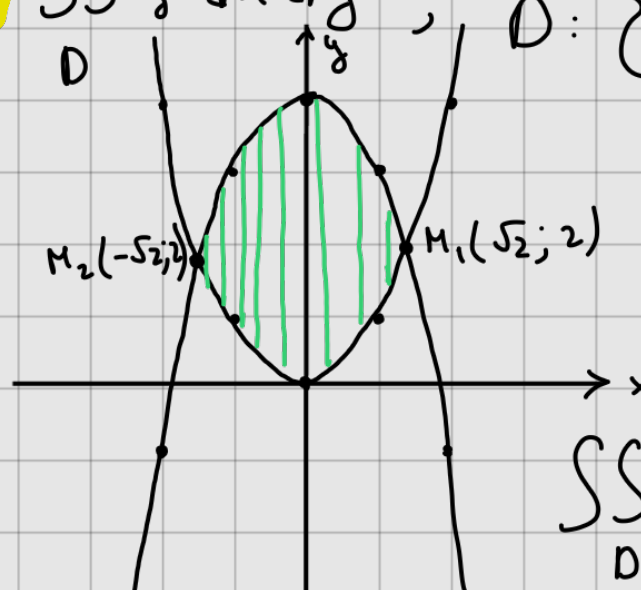
3. Двойные интегралы в произв. области

3.1. $\iint_D x dx dy$, D : $x+y=1$
 $x-y=1$
 $x=0$



$$\begin{aligned}
 \iint_D x dx dy &= \int_0^1 dx \int_{x-1}^{1-x} x dy = \\
 &= \int_0^1 \left(xy \Big|_{x-1}^{1-x} \right) dx = \\
 &= \int_0^1 (x(1-x) - x(x-1)) dx = \\
 &= \int_0^1 (x - x^2 - x^2 + x) dx = \int_0^1 (2x - 2x^2) dx = \\
 &= \left(x^2 - \frac{2x^3}{3} \right) \Big|_0^1 = 1 - \frac{2}{3} = \frac{1}{3}
 \end{aligned}$$

3.2. $\iint_D y \, dx \, dy$, $D: \begin{cases} y = x^2 \\ y = 4 - x^2 \end{cases}$

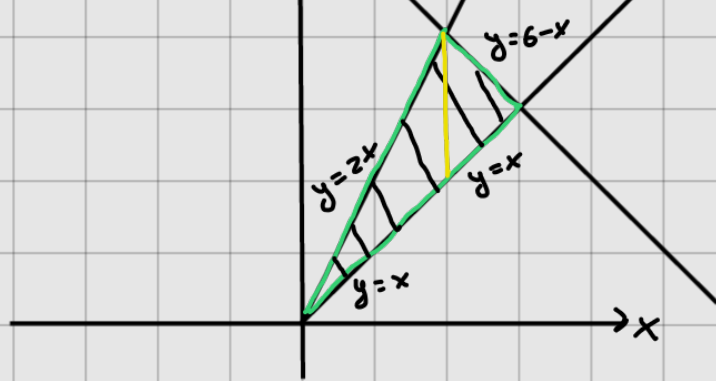


$x^2 = 4 - x^2$
 $2x^2 = 4$
 $x^2 = 2$
 $x = \pm\sqrt{2}$

$\iint_D y \, dx \, dy = \int_{-\sqrt{2}}^{\sqrt{2}} dx \int_{x^2}^{4-x^2} y \, dy =$
 $= \int_{-\sqrt{2}}^{\sqrt{2}} \left(\frac{y^2}{2} \Big|_{x^2}^{4-x^2} \right) dx = \int_{-\sqrt{2}}^{\sqrt{2}} \left(\frac{(4-x^2)^2}{2} - \frac{x^4}{2} \right) dx =$
 $= \int_{-\sqrt{2}}^{\sqrt{2}} \left(\frac{16 - 8x^2 + x^4}{2} - \frac{x^4}{2} \right) dx = \int_{-\sqrt{2}}^{\sqrt{2}} (8 - 4x^2) dx =$
 $= \left(8x - \frac{4}{3}x^3 \right) \Big|_{-\sqrt{2}}^{\sqrt{2}} = 8\sqrt{2} - \frac{4}{3}\sqrt{2^3} - \left(-8\sqrt{2} + \frac{4}{3}\sqrt{2^3} \right) =$
 $= 8\sqrt{2} - \frac{4}{3}\sqrt{8} + 8\sqrt{2} - \frac{4}{3}\sqrt{8} = 16\sqrt{2} - \frac{8}{3}\sqrt{8} =$
 $= 16\sqrt{2} - \frac{16\sqrt{2}}{3} = \frac{48\sqrt{2} - 16\sqrt{2}}{3} = \frac{32\sqrt{2}}{3}$

3.3.

$\iint_D dx \, dy$, $D: \begin{cases} y = x \\ y = 2x \\ x + y = 6 \end{cases}$



$$\iint_D dx dy = \int_0^2 dx \int_x^{2x} dy + \int_2^3 dx \int_x^{6-x} dy =$$

$$= \int_0^2 (y|_x^{2x}) dx + \int_2^3 (y|_x^{6-x}) dx = \int_0^2 (2x - x) dx + \int_2^3 (6 - x - x) dx =$$

$$= \frac{x^2}{2} \Big|_0^2 + 6x \Big|_2^3 - x^2 \Big|_2^3 = 2 + 18 - 12 - 9 + 4 = 3$$

3.4.

$$\iint_D y dx dy$$

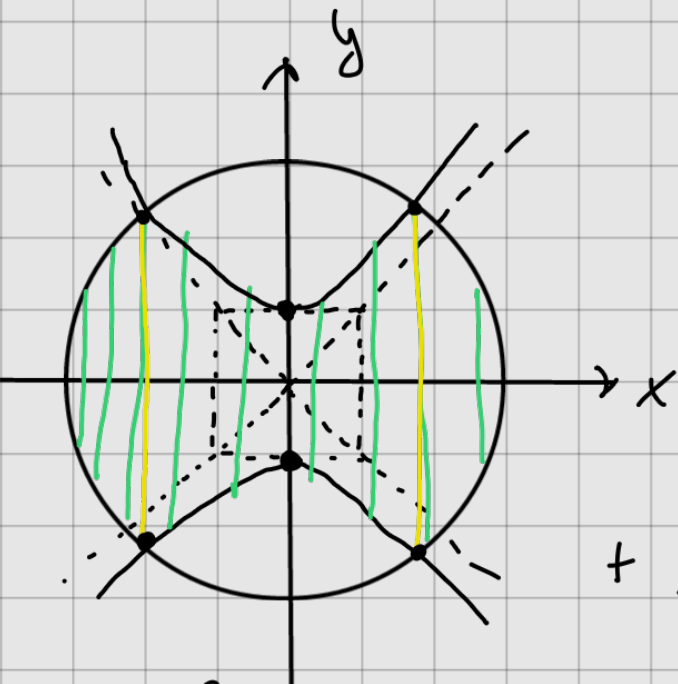
$$D: \begin{cases} y^2 - x^2 = 1 \Rightarrow y = \pm \sqrt{x^2 + 1} \\ x^2 + y^2 = 9 \Rightarrow y = \pm \sqrt{9 - x^2} \end{cases}$$

$$y^2 - x^2 - 1 = x^2 + y^2 - 9$$

$$2x^2 = 8$$

$$x = \pm 2$$

$$y = \pm \sqrt{9 - 4} = \pm \sqrt{5}$$



$$\iint_D y dx dy = 2 \left(\int_{-3}^{-2} dx \int_0^{\sqrt{9-x^2}} y dy + \int_{-2}^2 dx \int_0^{\sqrt{x^2+1}} y dy + \int_2^3 dx \int_0^{\sqrt{9-x^2}} y dy \right) =$$

$$= 2 \cdot \left(\int_{-3}^{-2} \left(\frac{y^2}{2} \Big|_0^{\sqrt{9-x^2}} \right) dx + \int_{-2}^2 \left(\frac{y^2}{2} \Big|_0^{\sqrt{x^2+1}} \right) dx + \int_2^3 \left(\frac{y^2}{2} \Big|_0^{\sqrt{9-x^2}} \right) dx \right) =$$

$$= 2 \cdot \left(\int_{-3}^{-2} \frac{9-x^2}{2} dx + \int_{-2}^2 \frac{x^2+1}{2} dx + \int_2^3 \frac{9-x^2}{2} dx \right) =$$

$$= \left(9x - \frac{x^3}{3} \right) \Big|_{-3}^{-2} + \left(\frac{x^3}{3} + x \right) \Big|_{-2}^2 + \left(9x - \frac{x^3}{3} \right) \Big|_2^3 =$$

$$= -18 + \frac{8}{3} - (-27 + 9) + \frac{8}{3} + 2 - \frac{8}{3} + 2 + 27 - 9 - 18 + \frac{8}{3} =$$

$$= -18 + \frac{8}{3} + 27 - 9 + \frac{8}{3} + 2 - \frac{8}{3} + 2 + 16 - 18 + \frac{8}{3} =$$

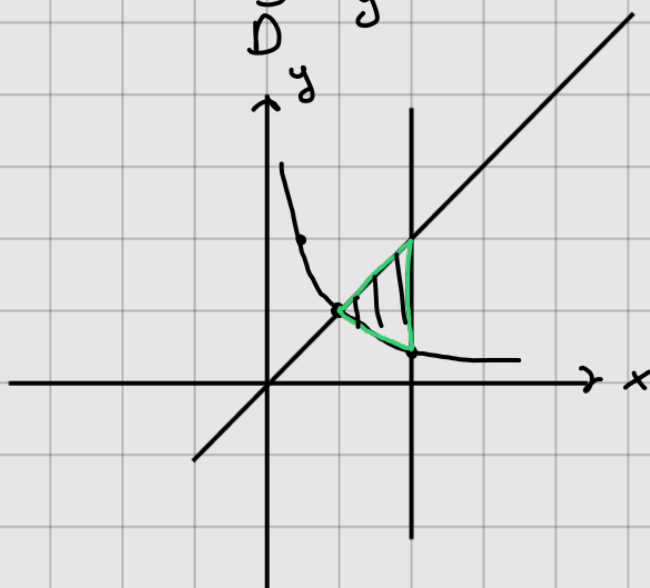
$$= 2 + \frac{16}{3} = \frac{22}{3} = 7 \frac{1}{3}$$

3.5.

$$\iint_D \frac{x^2}{y^2} dx dy,$$

$$D: x=2, y=x, xy=1$$

$$y = \frac{1}{x}$$



x	-2	-1	$-\frac{1}{2}$	$\frac{1}{2}$	1	2
y	$-\frac{1}{2}$	-1	-2	2	1	$\frac{1}{2}$

$$\iint_D \frac{x^2}{y^2} dx dy = \int_1^2 dx \int_{\frac{1}{x}}^x \frac{x^2}{y^2} dy =$$

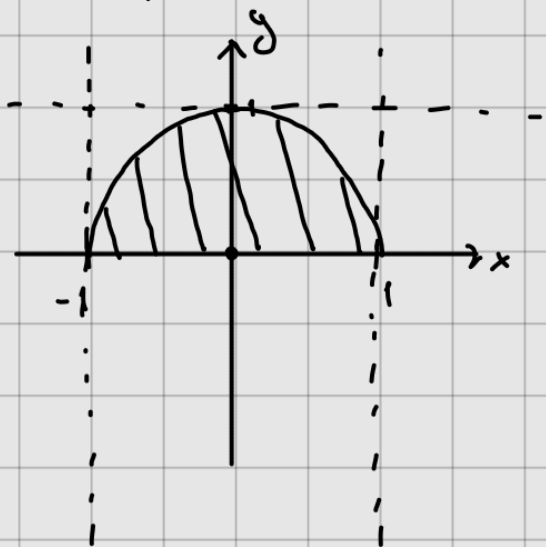
$$= \int_1^2 dx \int_{\frac{1}{x}}^x x^2 \cdot y^{-2} dy = \int_1^2 \left(-\frac{x^2}{y} \Big|_{\frac{1}{x}}^x \right) dx = \int_1^2 (x^3 - x) dx =$$

$$= \left(\frac{x^4}{4} - \frac{x^2}{2} \right) \Big|_1^2 = 4 - 2 - \frac{1}{4} + \frac{1}{2} = \frac{9}{4}$$

Двойные интегралы

1. Изменим порядок интегрирования

1.1.
$$\int_{-1}^1 dx \int_0^{\sqrt{1-x^2}} f dy = \int_0^1 dy \int_{-\sqrt{1-y^2}}^{\sqrt{1-y^2}} f dx$$



$$y = \sqrt{1-x^2}$$

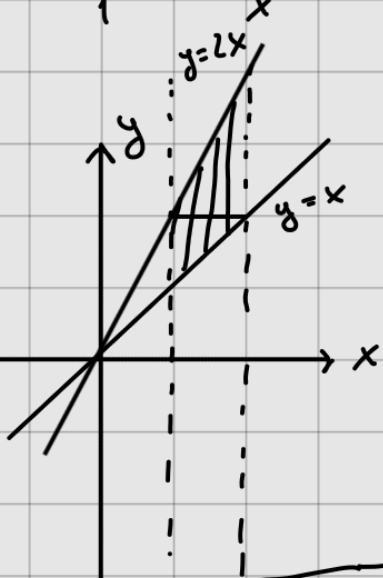
$$y^2 = 1-x^2$$

$$x^2 + y^2 = 1$$

$$x^2 = 1-y^2$$

$$x = \pm \sqrt{1-y^2}$$

1.2.
$$\int_1^2 dx \int_x^{2x} f(x,y) dy = \int_1^2 dy \int_1^y f(x,y) dx + \int_2^4 dy \int_{\frac{y}{2}}^2 f(x,y) dx$$



$$y = 2x$$

$$x = \frac{y}{2}$$

1.3.
$$\int_0^2 dy \int_y^{\sqrt{4y-y^2}} f(x,y) dx = \int_0^2 dx \int_{2-\sqrt{4-x^2}}^x f(x,y) dy$$



$$x = \sqrt{4y-y^2}$$

$$x^2 = 4y-y^2$$

$$x^2 + y^2 - 4y + 4 = 4$$

$$x^2 + (y-2)^2 = 2^2$$

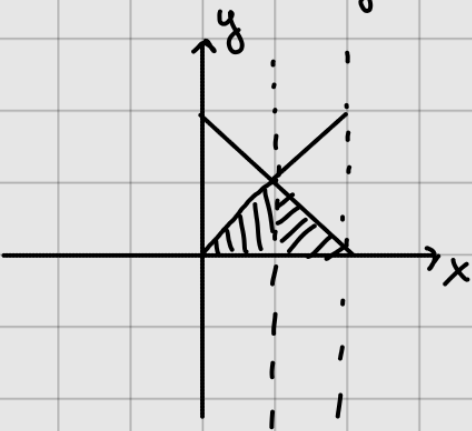
$$(y-2)^2 = 4 - x^2$$

$$y-2 = \pm \sqrt{4-x^2}$$

$$y = 2 - \sqrt{4-x^2}$$

1.4 $\int_0^1 dx \int_0^x f(x,y) dy + \int_1^2 dx \int_0^{2-x} f(x,y) dy =$

$$= \int_0^1 dy \int_y^{2-y} f(x,y) dx$$



$$y = x \Rightarrow x = y$$

$$y = 2 - x \Rightarrow x = 2 - y$$

2. Двойной интеграл в полярной системе координат.

2.1. $\iint_D xy \, dx \, dy$, $D: x^2 + y^2 \leq 2x, y \leq 0$

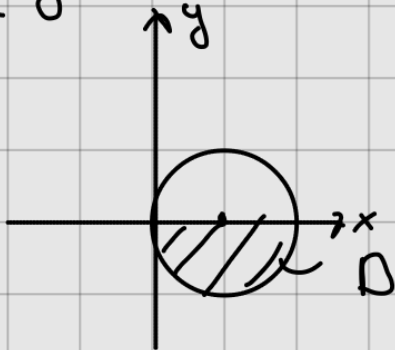
$$\begin{cases} x = \rho \cos \varphi \\ y = \rho \sin \varphi \end{cases}$$

$$I = \rho$$

$$x^2 - 2x + y^2 \leq 0$$

$$x^2 - 2x + 1 + y^2 \leq 1$$

$$(x-1)^2 + y^2 \leq 1^2$$



$$x^2 + y^2 = 2x$$

$$\rho^2 = 2\rho \cos \varphi$$

$$\rho = 2 \cos \varphi$$

$$\begin{aligned}
 \iint_D xy \, dx \, dy &= \int_{-\frac{\pi}{2}}^0 d\varphi \int_0^{2\cos\varphi} \rho \cos\varphi \rho \sin\varphi \cdot \rho \, d\rho \\
 &= \int_{-\frac{\pi}{2}}^0 \left(\cos\varphi \cdot \sin\varphi \cdot \frac{\rho^4}{4} \Big|_0^{2\cos\varphi} \right) d\varphi = \int_{-\frac{\pi}{2}}^0 \cos\varphi \sin\varphi \cdot 4\cos^4\varphi \, d\varphi = \\
 &= \int_{-\frac{\pi}{2}}^0 4\cos^5\varphi \sin\varphi \, d\varphi = -4 \int_{-\frac{\pi}{2}}^0 \cos^5\varphi \, d(\cos\varphi) = -4 \cdot \frac{\cos^6\varphi}{6} \Big|_{-\frac{\pi}{2}}^0 =
 \end{aligned}$$

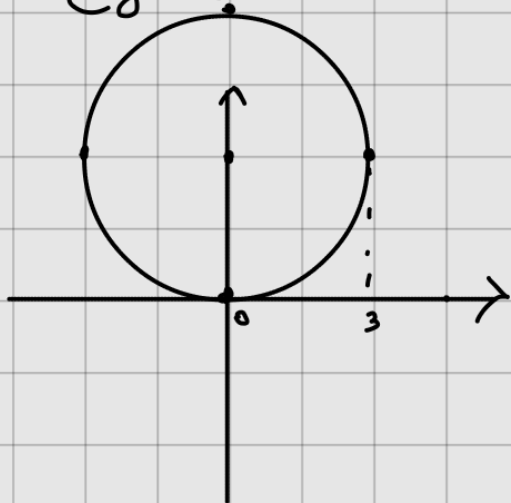
$$= -\frac{2}{3} \cos^6\varphi \Big|_{-\frac{\pi}{2}}^0 = -\frac{2}{3} (1 - 0) = -\frac{2}{3}$$

2.2. $\iint_D \sqrt{x^2 + y^2} \, dx \, dy$

$$D: x^2 + y^2 \leq 6y$$

$$\begin{cases} x = \rho \cos\varphi \\ y = \rho \sin\varphi \end{cases} \quad I = \rho$$

$$\begin{aligned}
 x^2 + y^2 - 6y + 9 &\leq 9 \\
 x^2 + (y - 3)^2 &\leq 3^2
 \end{aligned}$$



$$\begin{aligned}
 \iint_D \sqrt{x^2 + y^2} \, dx \, dy &= \int_0^{\pi} d\varphi \int_0^{6\sin\varphi} \sqrt{\rho^2 \cos^2\varphi + \rho^2 \sin^2\varphi} \cdot \rho \, d\rho = \\
 &= \int_0^{\pi} d\varphi \int_0^{6\sin\varphi} \rho^2 \, d\rho = \int_0^{\pi} \frac{\rho^3}{3} \Big|_0^{6\sin\varphi} d\varphi = \int_0^{\pi} 72 \sin^3\varphi \, d\varphi =
 \end{aligned}$$

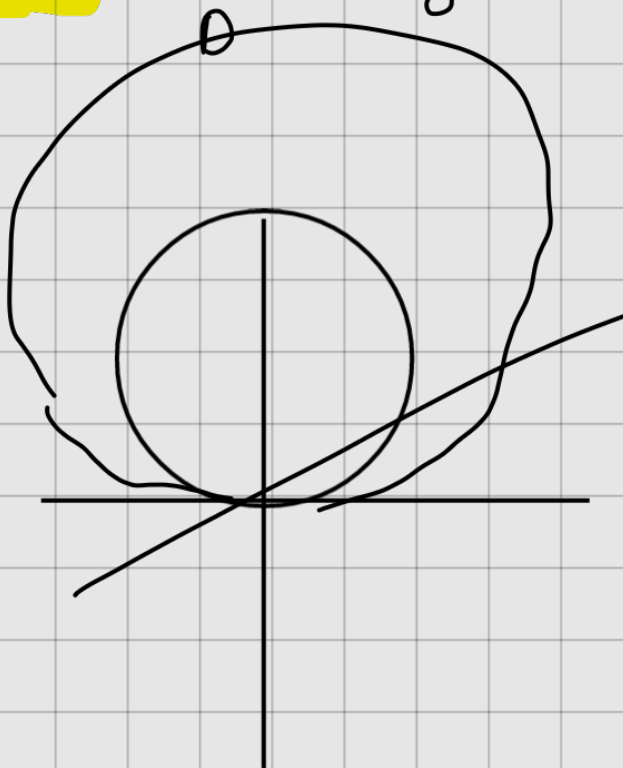
$$= 72 \int_0^{\pi} (1 - \cos^2 \varphi) \cdot \sin \varphi \, d\varphi = -72 \int_0^{\pi} (1 - \cos^2 \varphi) \, d(\cos \varphi) =$$

$$= \left[\begin{array}{l} \cos \varphi = u \\ du = -\sin \varphi \, d\varphi \\ \varphi_1 = \cos 0 = 1 \\ \varphi_2 = \cos \pi = -1 \end{array} \right] = -72 \int_1^{-1} (1 - u^2) \, du =$$

$$= 72 \int_{-1}^1 (1 - u^2) \, du = 72 \cdot \left(u - \frac{u^3}{3} \right) \Big|_{-1}^1 = 72 \cdot \left(1 - \frac{1}{3} \right) -$$

$$- 72 \cdot \left(-1 + \frac{1}{3} \right) = 72 \cdot \frac{2}{3} - 72 \cdot \left(-\frac{2}{3} \right) = 48 + 48 = 96$$

2.3. $\iint x \, dx \, dy$



$$\textcircled{D}: \begin{cases} y^2 - 4y + x^2 = 0 \\ y^2 - 8y + x^2 = 0 \\ y = \frac{x}{\sqrt{3}}, x=0 \end{cases}$$

$$\begin{cases} (y-2)^2 + x^2 = 4 \\ (y-4)^2 + x^2 = 16 \\ y = \frac{x}{\sqrt{3}}, x=0 \end{cases}$$

$$\begin{cases} x = \rho \cos \varphi \\ y = \rho \sin \varphi \end{cases}$$

D: verzeichneten C, ρ

$$\int_{\frac{\pi}{6}}^{\frac{\pi}{2}} d\varphi \int_{4\sin\varphi}^{8\sin\varphi} \rho \cos \varphi \, \rho \, d\rho$$

$$\begin{aligned} \rho &= 4\sin \varphi \\ \rho &= 8\sin \varphi \\ \varphi &= \frac{\pi}{2} \\ \varphi &= \frac{\pi}{6} \end{aligned}$$

3. Применение двойных интегралов

(площадь, масса плоской фигуры, статические моменты, центр масс, моменты инерции)

3.1. Площадь фигуры, охв. $y = \sqrt{x}$, $y = 2\sqrt{x}$,

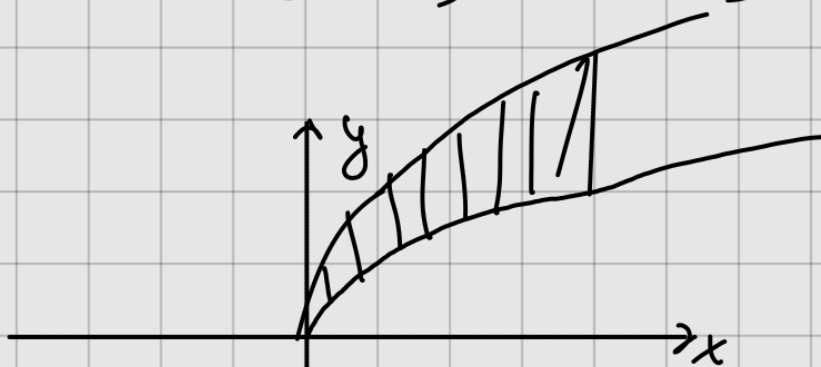
$$x = 4$$

$$S = \iint_D dx dy$$

$$S = \int_0^4 dx \int_{\sqrt{x}}^{2\sqrt{x}} dy =$$

$$= \int_0^4 y \Big|_{\sqrt{x}}^{2\sqrt{x}} dx = \int_0^4 (2\sqrt{x} - \sqrt{x}) dx = \int_0^4 \sqrt{x} dx = \frac{2}{3} x^{\frac{3}{2}} \Big|_0^4 =$$

$$= \frac{2}{3} \cdot 4^{\frac{3}{2}} = \frac{2}{3} \cdot 8 = \frac{16}{3} = 5 \frac{1}{3}$$



3.2. Фигура охв. $(x^2 + y^2)^2 = 2a^2(x^2 - y^2)$

$$\begin{cases} x = \rho \cos \varphi \\ y = \rho \sin \varphi \end{cases}$$

$$(\rho^2 \cos^2 \varphi + \rho^2 \sin^2 \varphi)^2 = 2a^2 (\rho^2 \cos^2 \varphi - \rho^2 \sin^2 \varphi)$$

$$\rho^4 = 2a^2 (\rho^2 \cos 2\varphi)$$

$$\rho^2 = 2a^2 \cdot \cos 2\varphi$$

$$\rho = a \sqrt{2 \cos 2\varphi}$$

$$S = 4 \int_0^{\frac{\pi}{4}} d\varphi \int_0^{a \sqrt{2 \cos 2\varphi}} \rho \, d\rho =$$

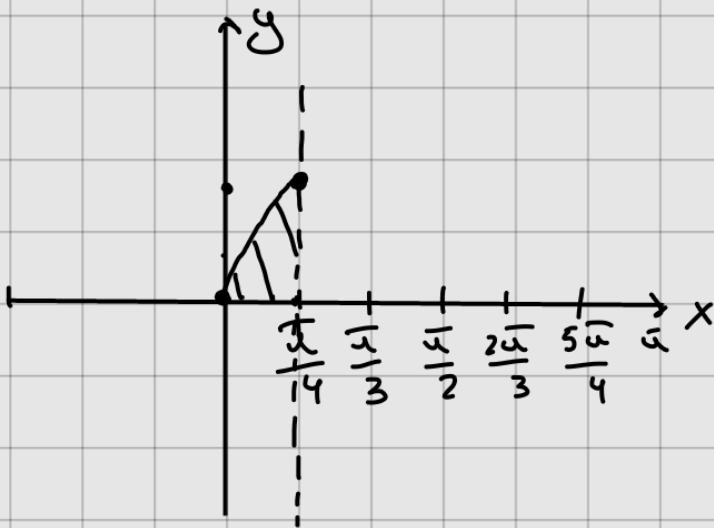
$$= 4 \int_0^{\frac{\pi}{4}} \frac{\rho^2}{2} \Big|_0^{a \sqrt{2 \cos 2\varphi}} d\varphi = 2 \int_0^{\frac{\pi}{4}} a^2 \cdot 2 \cos 2\varphi \, d\varphi =$$

$$= 4 a^2 \int_0^{\frac{\pi}{4}} \cos 2\varphi \, d\varphi = 4 a^2 \cdot \frac{1}{2} \cdot \sin 2\varphi \Big|_0^{\frac{\pi}{4}} = 2 a^2 \cdot (\sin \frac{\pi}{2} - \sin 0)$$

$$= 2 a^2$$

3.3. Найти центр масс фигуры, ор. $y = \sin x$,

$$y=0, \quad x = \frac{\pi}{4}$$



$$x_m = \frac{M_y}{m}$$

$$x_y = \frac{M_x}{m}$$

$$M_x = \iint_D y \cdot \delta(x, y) \, dx \, dy$$

$$m = \iint_D \delta(x, y) \, dx \, dy$$

$$m = \int_0^{\frac{\sqrt{2}}{2}} dx \int_0^{\sin x} dy = \int_0^{\frac{\sqrt{2}}{2}} \sin x dx = -\cos x \Big|_0^{\frac{\sqrt{2}}{2}} =$$

$$= -\frac{\sqrt{2}}{2} + 1 = 1 - \frac{\sqrt{2}}{2}$$

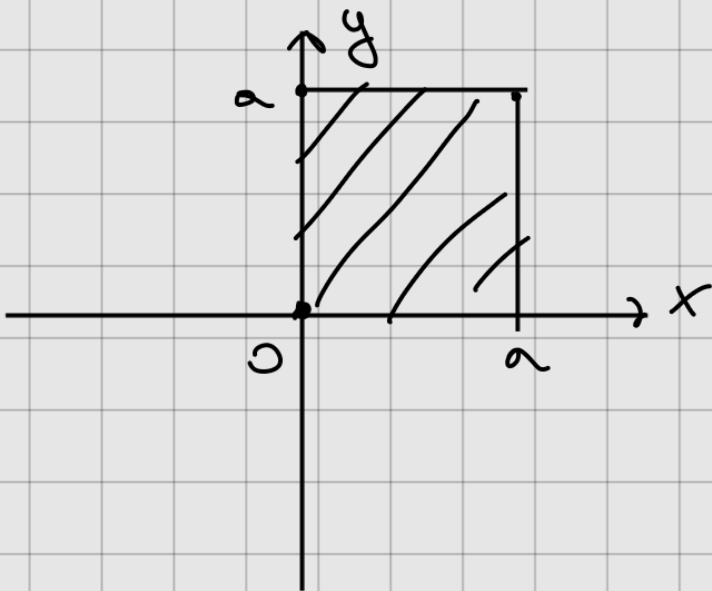
$$M_x = \iint_D y \cdot \delta(x, y) dx dy = \frac{\pi - 2}{16}$$

$$M_y = \iint_D x \cdot \delta(x, y) dx dy = \frac{\sqrt{2}}{2} \left(1 - \frac{\pi}{4}\right)$$

$$x_m = \frac{\frac{\sqrt{2}}{2} \left(1 - \frac{\pi}{4}\right)}{1 - \frac{\sqrt{2}}{2}}$$

$$y_m = \frac{\frac{\pi - 2}{16}}{1 - \frac{\sqrt{2}}{2}}$$

3.4. Момент инерции квадрата со стороной a отн. вершины

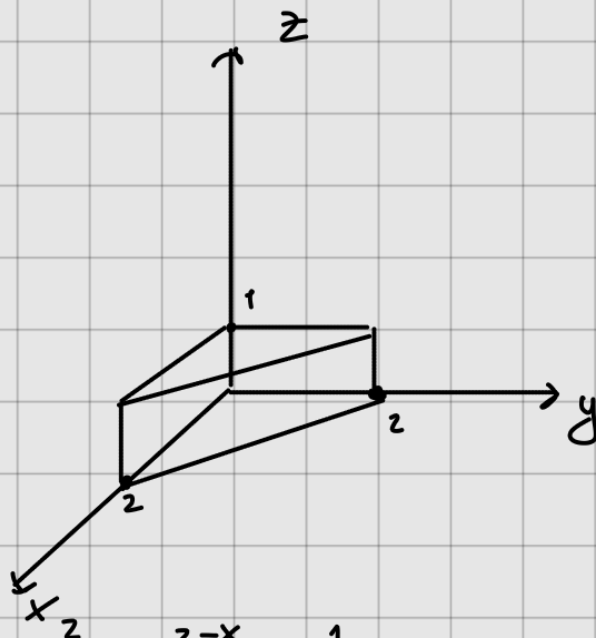
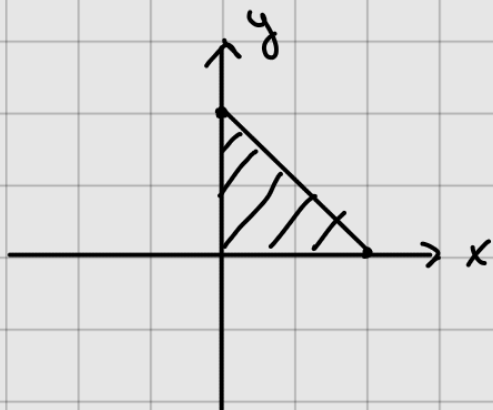


$$J_0 = \iint_D (x^2 + y^2) \delta(x, y) dx dy = \int_0^a dx \int_0^a (x^2 + y^2) dy = \frac{2a^3}{3}$$

Типовые интегралы

1. Двухпарное сечение укоротком

1.1. $\iiint_T (x+y-z) dx dy dz = ?$, $T: x=0, y=0, y=2-x, z=0, z=1$



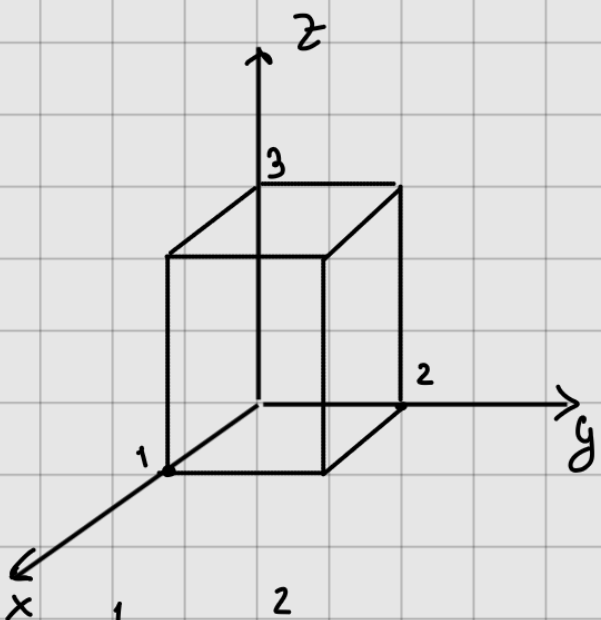
$$\begin{aligned}
 \iiint_T (x+y-z) dx dy dz &= \int_0^2 dx \int_0^{2-x} dy \int_0^1 (x+y-z) dz = \\
 &= \int_0^2 dx \int_0^{2-x} \left(xz + yz - \frac{z^2}{2} \right) \Big|_0^1 dy = \int_0^2 dx \int_0^{2-x} \left(x + y - \frac{1}{2} \right) dy = \\
 &= \int_0^2 \left(xy + \frac{y^2}{2} - \frac{1}{2}y \right) \Big|_0^{2-x} dx = \int_0^2 \left(2x - x^2 + \frac{4 - 4x + x^2}{2} - \frac{1}{2} \cdot (2-x) \right) dx = \\
 &= \int_0^2 \left(-\frac{x^2}{2} + \frac{x}{2} + 1 \right) dx = \left(-\frac{x^3}{6} + \frac{x^2}{4} + x \right) \Big|_0^2 =
 \end{aligned}$$

$$= -\frac{8}{6} + 1 + 2 = -\frac{4}{3} + \frac{9}{3} = \frac{5}{3}$$

1.2.

$$\iiint_T (x^2 + y^2 + z^2) dx dy dz$$

$$T: x=0, y=0, z=0, x=1, y=2, z=3$$



$$\iiint_T (x^2 + y^2 + z^2) dx dy dz =$$

$$= \int_0^1 dx \int_0^2 dy \int_0^3 (x^2 + y^2 + z^2) dz =$$

$$= \int_0^1 dx \int_0^2 \left(x^2 z + y^2 z + \frac{z^3}{3} \right) \Big|_0^3 dy =$$

$$= \int_0^1 dx \int_0^2 (3x^2 + 3y^2 + 9) dy = \int_0^1 (3x^2 y + y^3 + 9y) \Big|_0^2 dx =$$

$$= \int_0^1 (6x^2 + 8 + 18) dx = \int_0^1 (6x^2 + 26) dx = \frac{6x^3}{3} + 26x \Big|_0^1 =$$

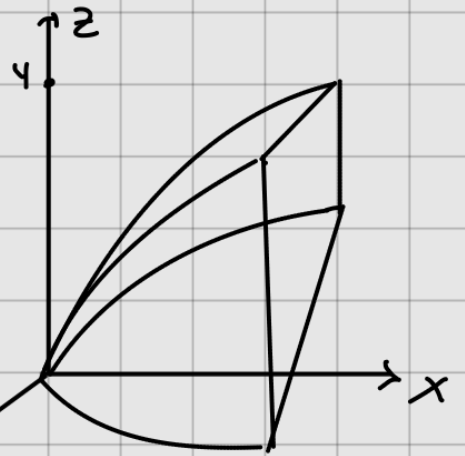
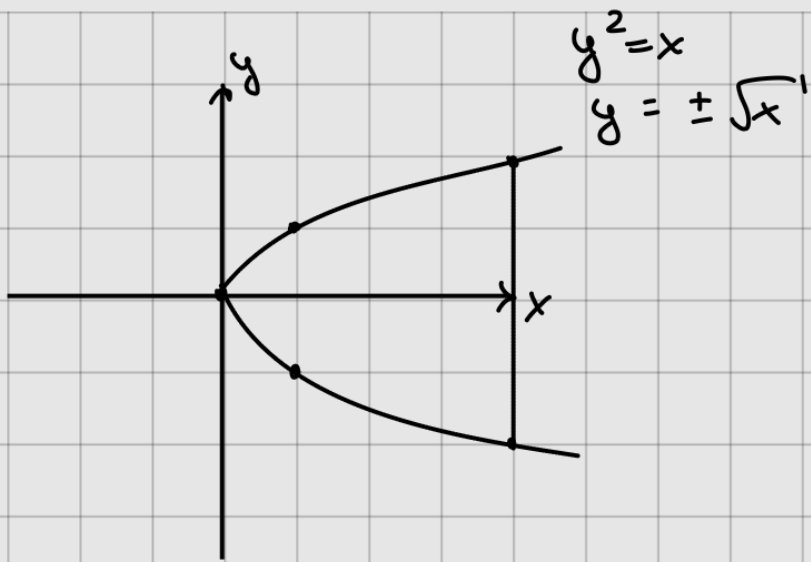
$$= 2 \cdot 1 + 26 = 28$$

1.3.

$$\iiint_T x dx dy dz = M_{yz}$$

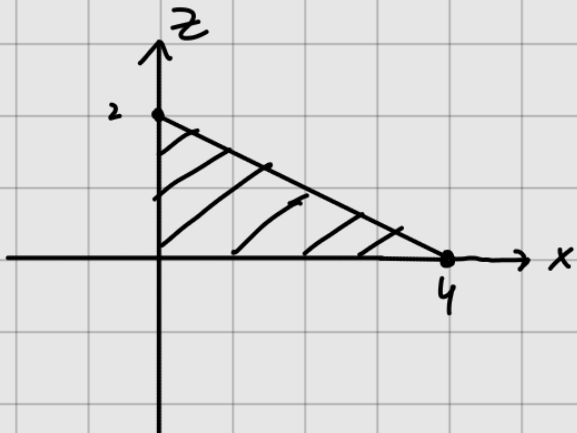
$$T: y^2 = x, x=4, z=0$$

$$z=x$$



$$\begin{aligned}
 \iiint x \, dx \, dy \, dz &= \int_0^4 dx \int_{-\sqrt{x}}^{\sqrt{x}} dy \int_0^x x \, dz = \\
 &= \int_0^4 dx \int_{-\sqrt{x}}^{\sqrt{x}} xz \Big|_0^x dy = \int_0^4 dx \int_{-\sqrt{x}}^{\sqrt{x}} x^2 dy = \\
 &= \int_0^4 x^2 y \Big|_{-\sqrt{x}}^{\sqrt{x}} dx = \int_0^4 x^2 \sqrt{x} - (-x^2 \sqrt{x}) dx = \int_0^4 (x^{\frac{5}{2}} + x^{\frac{5}{2}}) dx = \\
 &= \int_0^4 2x^{\frac{5}{2}} dx = 2 \cdot \frac{2}{7} x^{\frac{7}{2}} \Big|_0^4 = \frac{4}{7} \cdot 4^{\frac{7}{2}} = \frac{4}{7} \cdot 128 = \\
 &= \frac{512}{7}
 \end{aligned}$$

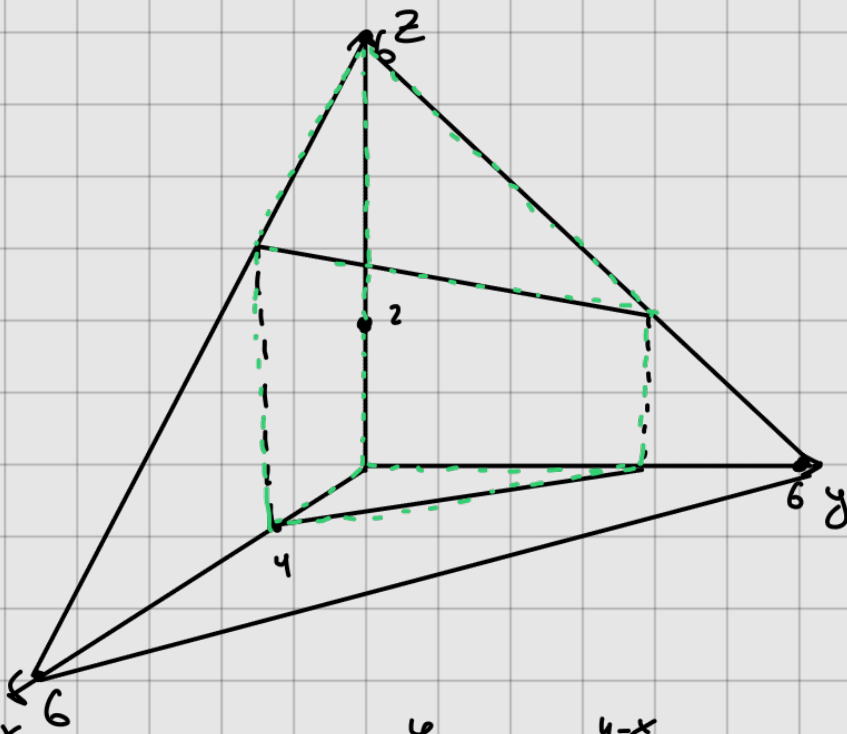
1.4. $T: x=0, y=0, z=0, x+2z=4, x+y+z=6$



$$x = 4 - 2z$$

$$y = 6 - x - z$$

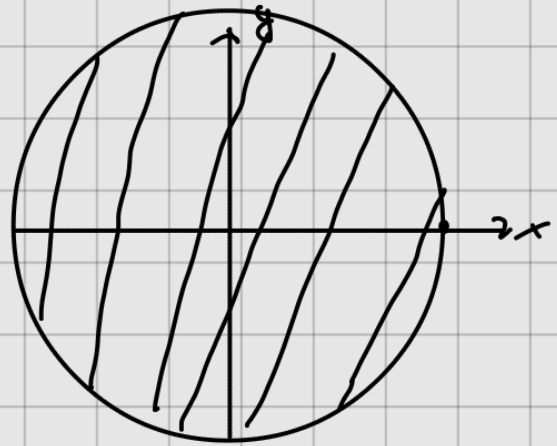
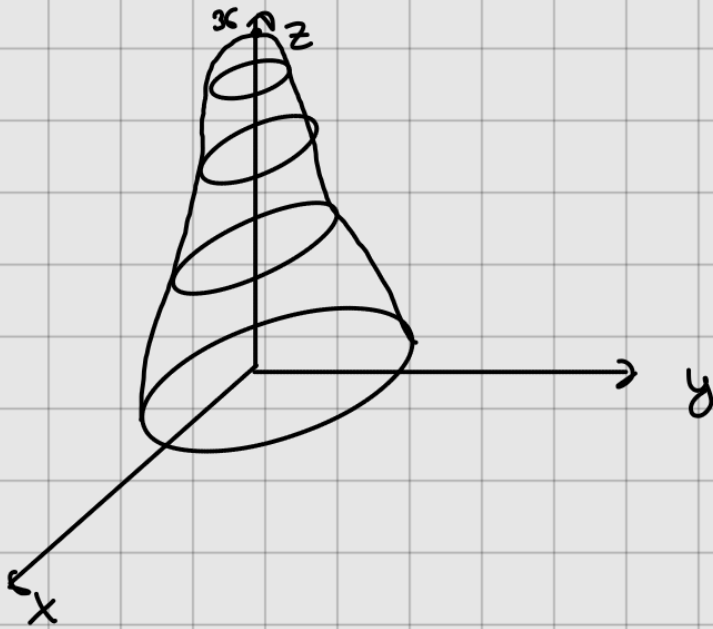
$$z = \frac{4-x}{2}$$



$$\begin{aligned}
 & \int_0^4 dx \int_0^{4-\frac{x}{2}} dz \int_0^{6-x-z} dy = \int_0^4 dx \int_0^{4-\frac{x}{2}} y \Big|_0^{6-x-z} dz = \\
 & = \int_0^4 dx \int_0^{4-\frac{x}{2}} (6-x-z) dz = \int_0^4 \left(6z - xz - \frac{z^2}{2} \right) \Big|_0^{4-\frac{x}{2}} dx = \\
 & = \int_0^4 \left(6 \left(4 - \frac{x}{2} \right) - x \cdot \left(4 - \frac{x}{2} \right) - \frac{\left(4 - \frac{x}{2} \right)^2}{2} \right) dx = \\
 & = \int_0^4 \left(24 - 3x - 4x + \frac{x^2}{2} - \frac{16 - 4x + \frac{x^2}{4}}{2} \right) dx = \\
 & = \int_0^4 \left(24 - 7x + \frac{x^2}{2} - 8 + 2x - \frac{x^2}{8} \right) dx = \\
 & = \int_0^4 \left(16 - 5x + \frac{3x^2}{8} \right) dx = \left(16x - \frac{5x^2}{2} + \frac{3x^3}{24} \right) \Big|_0^4 = \\
 & = 64 - \frac{5 \cdot 16}{2} + \frac{64}{8} = 64 - 40 + 8 = 32
 \end{aligned}$$

2. В цилиндрической системе координат

2.1. Найти объем тела, огр. $z=0$, $z=36-x^2-y^2$



$$\begin{aligned} z &= 36 - \rho \cos^2 \varphi - \rho \sin^2 \varphi = \\ &= 36 - \rho^2 \end{aligned}$$

Цилиндрические координаты

$$x = \rho \cos \varphi$$

$$y = \rho \sin \varphi$$

$$z = z$$

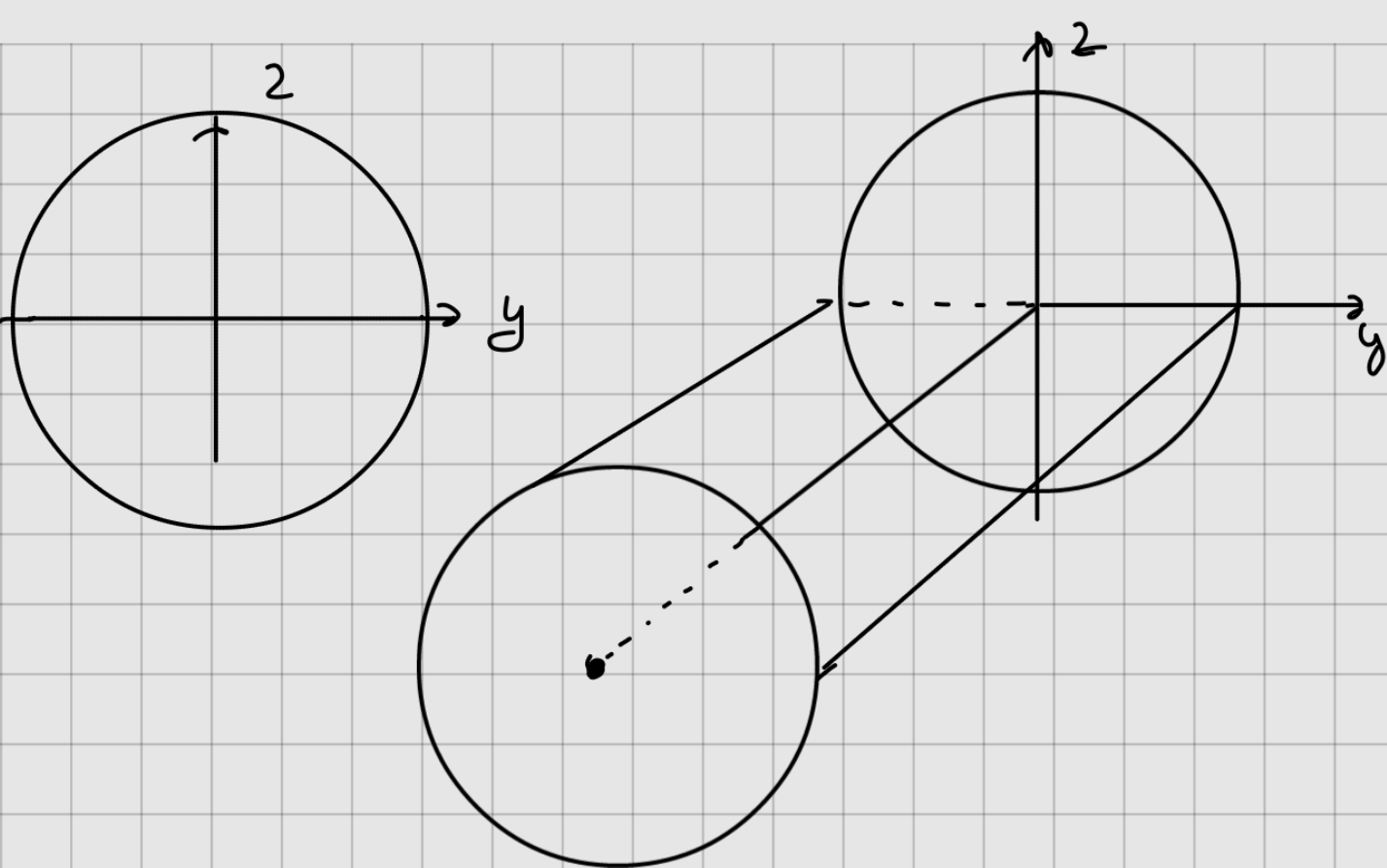
$$\begin{aligned} V &= \iiint_{T_6} dx dy dz = \int_0^{2\pi} d\varphi \int_0^6 \rho d\rho \int_0^{36-\rho^2} \rho dz = \\ &= 2\pi \int_0^6 \rho z \Big|_0^{36-\rho^2} d\rho = 2\pi \int_0^6 (36\rho - \rho^3) d\rho = \end{aligned}$$

$$= 2\pi \left(18\rho^2 - \frac{\rho^4}{4} \right) \Big|_0^6 = 2\pi \left(18 \cdot 36 - \frac{36 \cdot 36^2}{4} \right) =$$

$$= 2\pi \cdot 36 \cdot 9 = 36 \cdot 18\pi = 648\pi$$

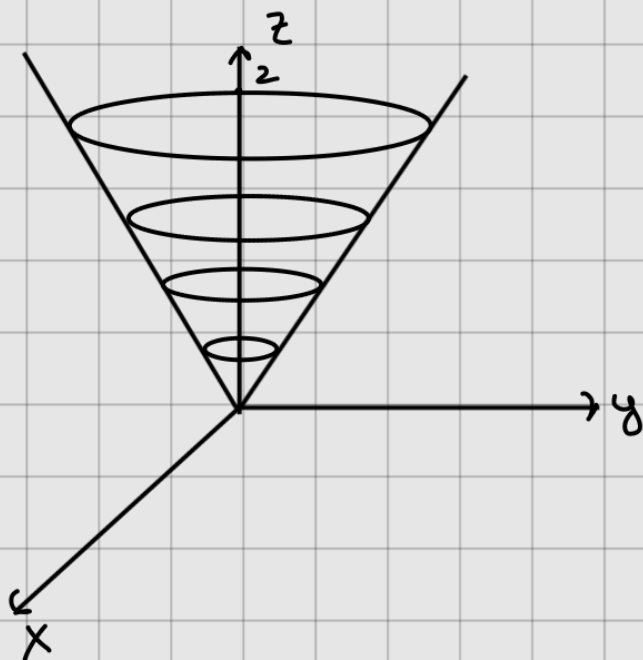
2.2. Найти момент инерции отн. оси Ox

тела, огр. $z^2 + y^2 = 9$, $x=z$, $x=5$, плоск. $y=x$



$$\begin{aligned}
 J_{0x} &= \iiint (y^2 + z^2) \cdot x \, dx \, dy \, dz = \\
 &= \int_0^{2\pi} d\varphi \int_0^3 dz \int_2^5 (\rho^2 \cdot x \cdot \rho) \, dx = 2\pi \cdot \int_0^3 \frac{\rho^3 x^2}{2} \Big|_2^5 \, d\rho = \\
 &= 2\pi \int_0^3 \left(\frac{25\rho^3}{2} - \frac{4\rho^3}{2} \right) d\rho = \pi \int_0^3 21\rho^3 \, d\rho = \\
 &= 21\pi \cdot \frac{\rho^4}{4} \Big|_0^3 = \frac{21\pi \cdot 81}{4} = \frac{1701\pi}{4}
 \end{aligned}$$

2.3. Найти массу тела $x^2 + y^2 = z^2$, $z=2$,
 $\rho = \frac{z}{2}$



$$m = \iiint_V \gamma(x, y, z) dx dy dz$$

$$x = \rho \cos \varphi \quad z = z$$

$$y = \rho \sin \varphi$$

$$I = \rho$$

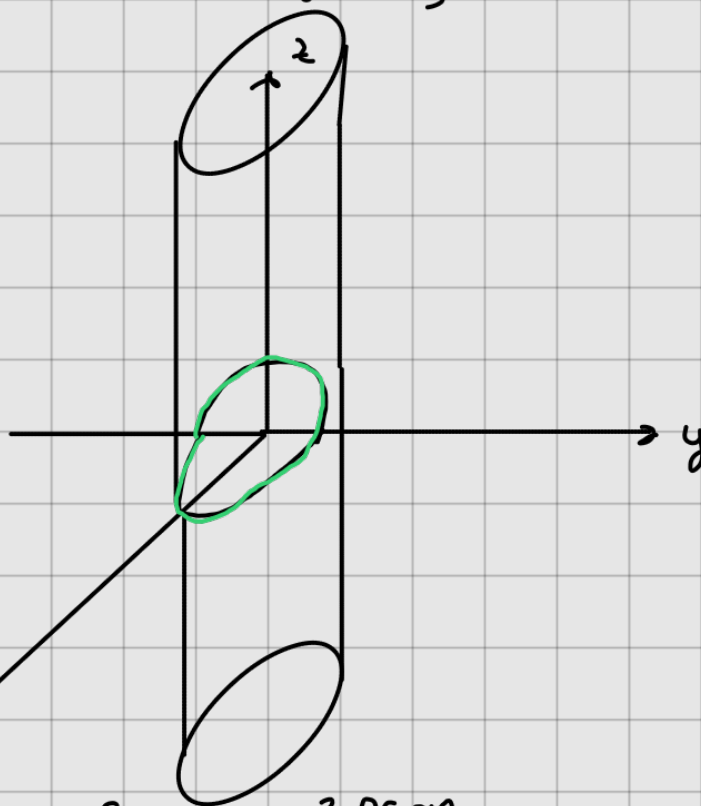
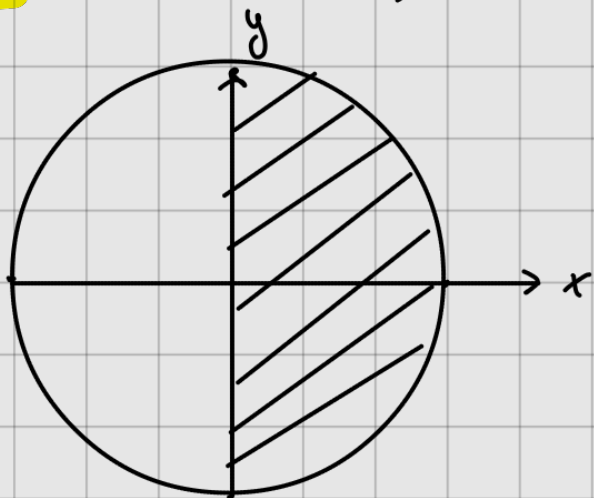
$$m = \iiint_V z dx dy dz = \int_0^{2\pi} d\varphi \int_0^2 dz \int_0^z z \cdot \rho d\rho =$$

$$= \int_0^{2\pi} d\varphi \int_0^2 \frac{z \rho^2}{2} \Big|_0^z dz = \int_0^{2\pi} d\varphi \int_0^2 \frac{z^3}{2} dz =$$

$$= \int_0^{2\pi} \frac{z^4}{8} \Big|_0^2 d\varphi = 2 \cdot 2\pi = 4\pi$$

Плоские интегралы (применение)

1. Найти V , огр. $z=0$, $z=2x$, $x^2+y^2=9$, $x \geq 0$



$$V = \iiint_T dx dy dz = \int_{-\pi/2}^{\pi/2} d\varphi \int_0^3 \rho d\rho \int_0^{2\rho \cos \varphi} dz =$$

$$x = \rho \cos \varphi$$

$$y = \rho \sin \varphi$$

$$z = z$$

$$= \int_{-\pi/2}^{\pi/2} d\varphi \int_0^3 \rho z \Big|_0^{2\rho \cos \varphi} d\rho =$$

$$= 2 \int_{-\pi/2}^{\pi/2} d\varphi \int_0^3 \rho^2 \cos \varphi d\rho = 2 \int_{-\pi/2}^{\pi/2} \frac{\rho^3}{3} \cos \varphi d\rho \Big|_0^3 d\varphi =$$

$$= 18 \left(\sin \frac{\varphi}{2} - \sin(-\frac{\varphi}{2}) \right) = 18 \cdot (1 + 1) = 18 \cdot 2 = 36$$

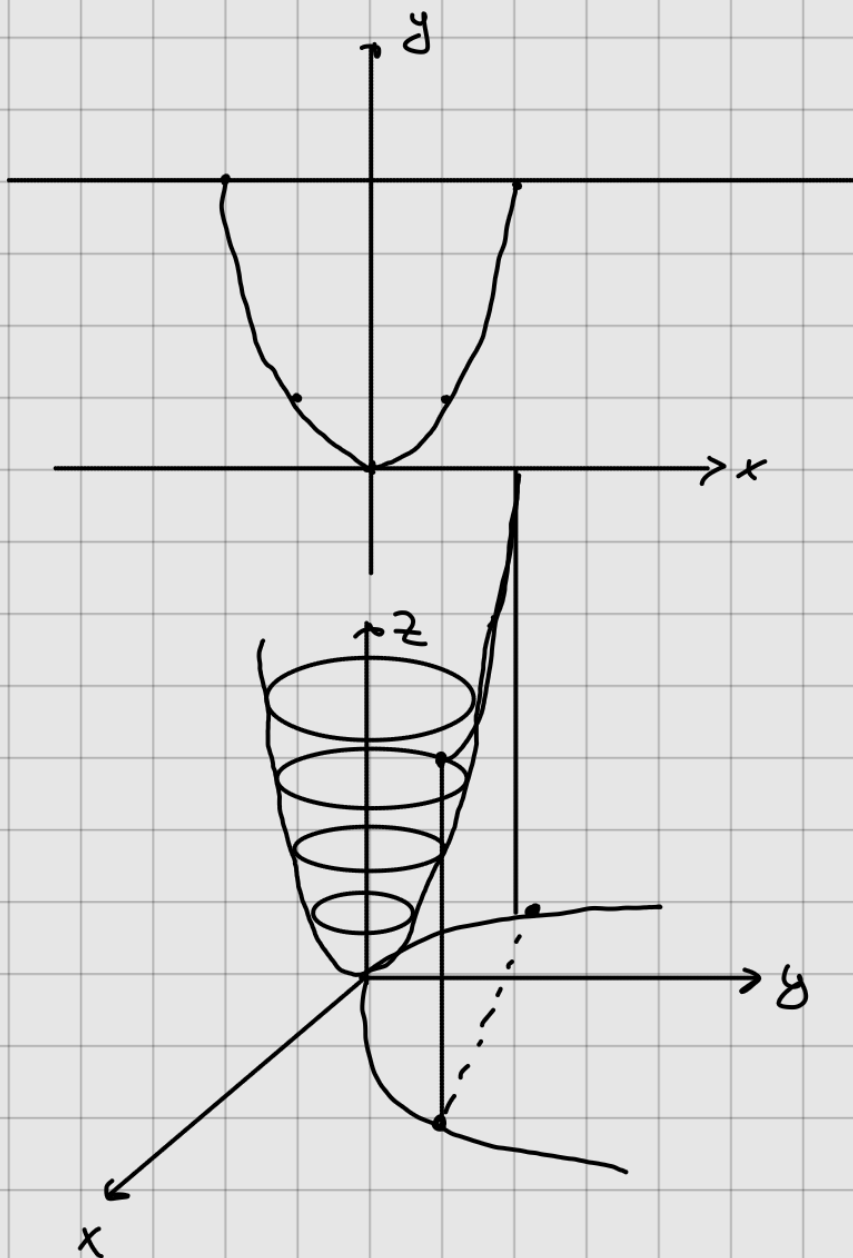
Другой способ: $V = \int_0^3 dx \int_{-\sqrt{9-x^2}}^{\sqrt{9-x^2}} dy \int_0^{2x} dz =$

$$= \int_0^3 dx \int_{-\sqrt{9-x^2}}^{\sqrt{9-x^2}} 2x dy = 2 \int_0^3 xy \Big|_{-\sqrt{9-x^2}}^{\sqrt{9-x^2}} dx = 2 \int_0^3 2x \sqrt{9-x^2} dx =$$

$$= -2 \int_0^3 \sqrt{9-x^2} d(9-x^2) = -2 \frac{(9-x^2)^{\frac{3}{2}}}{\frac{3}{2}} = -\frac{4}{3} (0 - 27) =$$

$$= 36$$

2. $T: z=0, y=4, y=x^2, z=x^2+y^2$



$$V = \int_{-2}^2 dx \int_{x^2}^4 dy \int_0^{x^2+y^2} dz = \int_{-2}^2 dx \int_{x^2}^4 (x^2+y^2) dy =$$

$$\begin{aligned}
 &= \int_{-2}^2 \left(x^2 y + \frac{y^3}{3} \right) \bigg|_{x^2}^4 dx = \int_{-2}^2 \left(4x^2 + \frac{64}{3} - x^4 - \frac{x^6}{3} \right) dx = \\
 &= \left(\frac{4x^3}{3} + \frac{64x}{3} - \frac{x^5}{5} - \frac{x^7}{7} \right) \bigg|_{-2}^2 = \frac{6016}{105}
 \end{aligned}$$

Криволинейные и поверхностные интегралы

1. Найти массу $y = \ln x$ с плотностью $\delta = x^2$

от $x_1 = 1$ до $x_2 = 4$

$$m = \int_L \delta(x, y) dL = \int_1^4 x^2 \sqrt{1 + \frac{1}{x^2}} dx = \int_1^4 x^2 \sqrt{\frac{x^2 + 1}{x^2}} dx \quad \textcircled{=}$$

$$dL = \sqrt{1 + (y')^2} dx$$

$$\textcircled{=} \int_1^4 x \sqrt{x^2 + 1} dx = \frac{1}{2} \int_1^4 \sqrt{x^2 + 1} d(x^2 + 1) = \frac{1}{2} (x^2 + 1)^{\frac{3}{2}} \cdot \frac{2}{3} \Big|_1^4$$

$$= \frac{1}{3} \cdot \left(17^{\frac{3}{2}} - 2^{\frac{3}{2}} \right) \approx 21,1$$

2. Вычислить $\int_L \frac{z^2 dL}{x^2 + y^2}$, L — винтовая линия Бернулли, $x = a \cos t$, $y = a \sin t$, $z = at$

$$\begin{aligned} \int_L \frac{z^2 dL}{x^2 + y^2} &= \int_0^{2\pi} \frac{a^2 t^2 dt}{a^2 \cos^2 t + a^2 \sin^2 t} \cdot \sqrt{a^2 \cos^2 t + a^2 \sin^2 t + a^2} dt = \\ &= \int_0^{2\pi} t^2 \cdot a \sqrt{2} dt = a \sqrt{2} \cdot \frac{1}{3} t^3 \Big|_0^{2\pi} = \frac{8\pi^3 \cdot a \sqrt{2}}{3} \end{aligned}$$

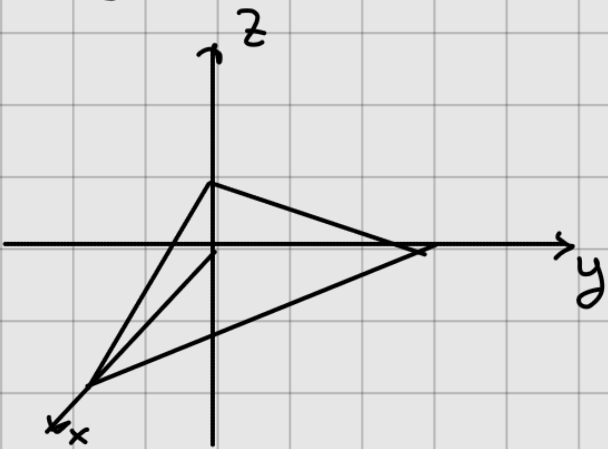
3. $\int_L \arctg \frac{y}{x} dL$, L - часть кривой спирали $\rho = 2\varphi$
 $0 \leq \varphi \leq \frac{\pi}{4}$

$$\begin{aligned} \int_L \arctg \frac{y}{x} dL &= \int_0^{\frac{\pi}{4}} \arctg \frac{\rho \sin \varphi}{\rho \cos \varphi} \cdot \sqrt{\rho^2 + (\rho')^2} d\varphi = \\ &= \int_0^{\frac{\pi}{4}} \arctg(\tg \varphi) \cdot \sqrt{4\rho^2 + 4} d\varphi = \int_0^{\frac{\pi}{4}} \varphi \cdot 2\sqrt{\varphi^2 + 1} d\varphi = \\ &= \int_0^{\frac{\pi}{4}} \sqrt{\varphi^2 + 1} d(\varphi^2 + 1) = \left(\varphi^2 + 1 \right)^{\frac{3}{2}} \cdot \frac{2}{3} \Big|_0^{\frac{\pi}{4}} = \\ &= \frac{2}{3} \cdot \left(\left(\frac{\pi^2}{16} + 1 \right)^{\frac{3}{2}} - 1 \right) \end{aligned}$$

4. $\iint_{\sigma} z \cdot d\sigma$ $\sigma: x + 2y + 3z = 6$ 1-ая окт.
 $z = \frac{6 - x - 2y}{3} = 2 - \frac{x}{3} - \frac{2}{3}y$

$$\iint_{\sigma} z \cdot d\sigma = \iint_{\sigma} z \cdot \sqrt{1 + \left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2} dx dy =$$

$$\begin{aligned} &= \iint_{\sigma} z \cdot \sqrt{1 + \frac{1}{9} + \frac{4}{9}} dx dy = \iint_{\sigma} \left(2 - \frac{x}{3} - \frac{2}{3}y \right) \cdot \frac{\sqrt{14}}{3} dx dy = \\ &= \int_0^6 dx \int_0^{3 - \frac{1}{2}x} \left(2 - \frac{1}{3}x - \frac{2}{3}y \right) \cdot \frac{\sqrt{14}}{3} dy \quad \textcircled{=} \end{aligned}$$



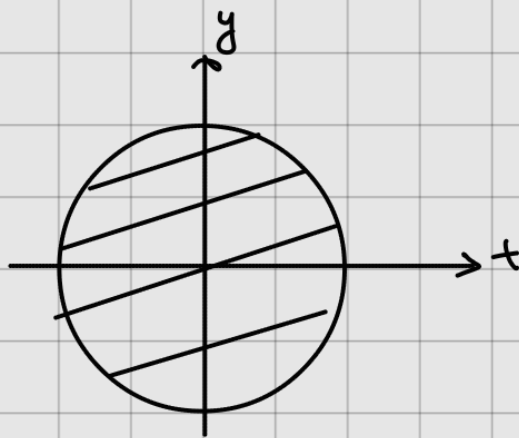
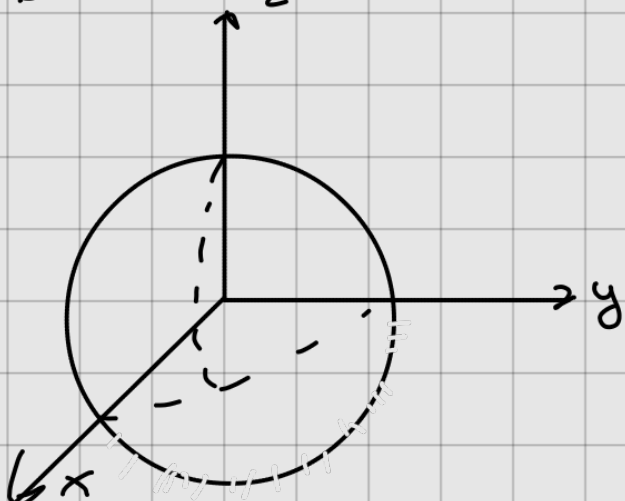
$$\begin{aligned} z = 0: \quad \frac{2}{3}y &= 2 - \frac{x}{3} \\ y &= 3 - \frac{x}{2} \end{aligned}$$

$$y = 0: x = 6$$

$$\begin{aligned}
 & \textcircled{=} \frac{\sqrt{14}}{3} \int_0^6 \left(2y - \frac{1}{3}xy - \frac{1}{3}y^2 \right) \Big|_0^{3 - \frac{1}{2}x} dx = \\
 & = \frac{\sqrt{14}}{3} \int_0^6 \left(6 - x - x + \frac{1}{6}x^2 - \frac{1}{3} \left(9 - 3x + \frac{1}{4}x^2 \right) \right) dx = \\
 & = \frac{\sqrt{14}}{3} \int_0^6 \left(3 - x + \frac{1}{12}x^2 \right) dx = \frac{\sqrt{14}}{3} \cdot \left(3x - \frac{x^2}{2} + \frac{x^3}{36} \right) \Big|_0^6 = \\
 & = \frac{\sqrt{14}}{3} \cdot (18 - 18 + 6) = 2\sqrt{14}
 \end{aligned}$$

5. $\iint_{\delta} y \, d\sigma \textcircled{=} \quad \delta: z = \sqrt{R^2 - x^2 - y^2}$

$$\begin{aligned}
 \textcircled{=} \iint_D \sqrt{1 + \left(\frac{\partial z}{\partial x} \right)^2 + \left(\frac{\partial z}{\partial y} \right)^2} \, dx \, dy &= \iint_D y \sqrt{1 + \left(\frac{-x}{\sqrt{R^2 - y^2 - x^2}} \right)^2 +} \\
 &+ \left(\frac{-y}{\sqrt{R^2 - y^2 - x^2}} \right)^2} \, dx \, dy = \iint_D y \cdot \sqrt{1 + \frac{x^2 + y^2}{R^2 - y^2 - x^2}} \, dx \, dy = \\
 &= \iint_D y \sqrt{\frac{R^2}{R^2 - y^2 - x^2}} \, dx \, dy = R \iint_D y \sqrt{\frac{1}{R^2 - x^2 - y^2}} \, dx \, dy
 \end{aligned}$$



$$\textcircled{=} R \int_0^{2\pi} d\varphi \int_0^R \rho \sin \varphi \cdot \sqrt{\frac{1}{R^2 - \rho^2}} \cdot \rho d\rho =$$

$$= R \int_0^{2\pi} \sin \varphi d\varphi \int_0^R \rho^2 \cdot \sqrt{\frac{1}{R^2 - \rho^2}} d\rho = R \int_0^R \rho^2 \frac{1}{\sqrt{R^2 - \rho^2}} d\rho \int_0^{2\pi} \sin \varphi d\varphi =$$

$$= R \int_0^{2\pi} \sin \varphi d\varphi \int_0^R \rho^2 \cdot \frac{1}{\sqrt{R^2 - \rho^2}} \cdot (-\cos \varphi) \Big|_0^{2\pi} d\rho = 0$$

Числовые ряды

1. Усложнение на сходимость

1.1.

$$\sum_{n=1}^{\infty} \frac{n^4}{(3n^2+3)^3}$$

$$a_n = \frac{n^4}{(3n^2+3)^3} \sim \frac{n^4}{3n^6} = \frac{1}{n^2} = b_n$$

$$\sum_{n=1}^{\infty} b_n = \sum_{n=1}^{\infty} \frac{1}{n^2} - \text{сходящееся } (p > 1; n^p)$$

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \lim_{n \rightarrow \infty} \frac{n^4}{(3n^2+3)^3} \cdot \frac{n^2}{1} = \lim_{n \rightarrow \infty} \frac{n^6}{(3n^2+3)^3} =$$

$$= \lim_{n \rightarrow \infty} \frac{\cancel{n^6}}{(\cancel{n^6} (3 + \frac{3}{n^2})^3)} = \lim_{n \rightarrow \infty} \frac{1}{(3 + \frac{3}{n^2})^3} = \frac{1}{27} \neq 0 \Rightarrow \text{сход.}$$

1.2.

$$\sum_{n=1}^{\infty} \frac{n+4}{n^2+2n-2}$$

$$a_n = \frac{n+4}{n^2+2n-2} \sim \frac{n}{n^2} = \frac{1}{n} = b_n$$

$$\sum_{n=1}^{\infty} b_n = \sum_{n=1}^{\infty} \frac{1}{n} - \text{расходящееся (гармонический, } p=1)$$

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \lim_{n \rightarrow \infty} \frac{n+4}{n^2+2n-2} \cdot \frac{n}{1} = \lim_{n \rightarrow \infty} \frac{n^2+4n}{n^2+2n-2} =$$

$$= \lim_{n \rightarrow \infty} \frac{\cancel{n^2} (1 + \frac{4}{n})}{\cancel{n^2} (1 + \frac{2}{n} - \frac{2}{n^2})} = \frac{1}{1} = 1 \neq 0 \Rightarrow \text{расходящееся}$$

1.3.

$$\sum_{n=1}^{\infty} \frac{(n+2)^2}{4+2n+n^3}$$

$$a_n = \frac{(n+2)^2}{4+2n+n^3} \sim \frac{n^2}{n^3} = \frac{1}{n} = b_n$$

$$\sum_{n=1}^{\infty} b_n = \sum_{n=1}^{\infty} \frac{1}{n} \text{ - расходящееся (гармонический ряд)}$$

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \lim_{n \rightarrow \infty} \frac{(n+2)^2}{4+2n+n^3} \cdot \frac{n}{1} = \lim_{n \rightarrow \infty} \frac{n^2+2n+4}{4+2n+n^3} \cdot \frac{n}{1} =$$

$$= \lim_{n \rightarrow \infty} \frac{n^3+2n^2+4n}{4+2n+n^3} = \lim_{n \rightarrow \infty} \frac{n^3(1+\frac{2}{n}+\frac{4}{n^2})}{n^3(\frac{4}{n^3}+\frac{2}{n^2}+1)} =$$

$$= \frac{1}{1} = 1 \neq 0 \text{ - расходящееся}$$

1.4.

$$\sum_{n=1}^{\infty} \frac{(\sqrt{n}+2)^2}{n(1+2n^2)}$$

$$a_n = \frac{(\sqrt{n}+2)^2}{n(1+2n^2)} = \frac{n+4\sqrt{n}+4}{n+2n^3} \sim \frac{n}{n^3} = \frac{1}{n^2} = b_n$$

$$= b_n$$

$$\sum_{n=1}^{\infty} b_n = \sum_{n=1}^{\infty} \frac{1}{n^2} \text{ - сходящееся (p > 1)}$$

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \lim_{n \rightarrow \infty} \frac{n+4\sqrt{n}+4}{n+2n^3} \cdot \frac{n^2}{1} = \frac{n^3+4n^{\frac{5}{2}}+4n^2}{n+2n^3} =$$

$$= \lim_{n \rightarrow \infty} \frac{n^3 \left(1 + \frac{4}{\sqrt{n}} + \frac{4}{n}\right)}{n^3 \left(\frac{1}{n^2} + 2\right)} = \frac{1}{2} \Rightarrow \text{сходящаяся}$$

2. Признак Д'Aламбера

Если $\exists L$ и $L < 1$ - сходя.

$L > 1$ - расходя.

$L = 1$ - признак неясен

2.1. $\sum_{n=1}^{\infty} \frac{5n+1}{3^n}$

$$a_n = \frac{5n+1}{3^n}$$

$$a_{n+1} = \frac{5(n+1)+1}{3^{n+1}} = \frac{5n+6}{3 \cdot 3^n}$$

$$L = \lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \lim_{n \rightarrow \infty} \frac{5n+6}{3 \cdot 3^n} \cdot \frac{3^n}{5n+1} = \lim_{n \rightarrow \infty} \frac{n(5 + \frac{6}{n})}{15n + 3} =$$

$$= \lim_{n \rightarrow \infty} \frac{n(5 + \frac{6}{n})}{n(15 + \frac{3}{n})} = \frac{5}{15} = \frac{1}{3} < 1 - \text{сходящаяся}$$

2.2. $\sum_{n=1}^{\infty} \frac{2^{n+1}}{(4n)!}$

$$a_n = \frac{2^{n+1}}{(4n)!} = \frac{2 \cdot 2^n}{(4n)!}$$

$$a_{n+1} = \frac{2 \cdot 2^{n+1}}{(4(n+1))!} = \frac{4 \cdot 2^n}{(4n+4)!}$$

$$L = \lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \lim_{n \rightarrow \infty} \frac{4 \cdot 2^n}{(4n+4)!} \cdot \frac{(4n)!}{2 \cdot 2^n} =$$

$$= \lim_{n \rightarrow \infty} \frac{2 \cdot (4n)!}{(4n)! \cdot (4n+1) \cdot (4n+2) \cdot (4n+3) \cdot (4n+4)} = \lim_{n \rightarrow \infty} \frac{2}{(4n+1)(4n+2)(4n+3)(4n+4)} =$$

$$= \frac{2}{\infty} = 0 < 1 - \text{сходимость}$$

$$2.3. \sum_{n=1}^{\infty} \frac{n^2 + 5n}{4^{2n+1}} - \text{сход.}$$

$$a_n = \frac{n^2 + 5n}{4^{2n+1}}$$

$$a_{n+1} = \frac{n^2 + 2n + 1 + 5(n+1)}{4^{2(n+1)+1}} = \frac{n^2 + 7n + 6}{4^{2n+3}} = \frac{n^2 + 7n + 6}{64 \cdot 4^{2n}}$$

$$L = \lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \lim_{n \rightarrow \infty} \frac{n^2 + 7n + 6}{\cancel{64} \cdot \cancel{4^{2n}} \cdot \frac{16}{16}} \cdot \frac{\cancel{4 \cdot 4^{2n}}}{n^2 + 5n} =$$

$$= \lim_{n \rightarrow \infty} \frac{\cancel{n^2} \left(1 + \frac{7}{n} + \frac{16}{n^2}\right)}{16 \cancel{n^2} \left(1 + \frac{5}{n}\right)} = \frac{1}{16} < 1 \Rightarrow \text{сходимость}$$

$$2.4. \sum_{n=1}^{\infty} \frac{3^{n+1}}{(2n+1)!}$$

$$a_n = \frac{3^{n+1}}{(2n+1)!} = \frac{3 \cdot 3^n}{(2n+1)!}$$

$$a_{n+1} = \frac{3^{n+2}}{(2(n+1)+1)!} = \frac{9 \cdot 3^n}{(2n+3)!} = \frac{9 \cdot 3^n}{(2n+2) \cdot (2n+3) \cdot (2n+1)!}$$

$$L = \lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \frac{\cancel{9} \cdot \cancel{3} \cdot 3}{(2n+2)(2n+3) \cdot \cancel{(2n+1)!}} \cdot \frac{\cancel{(2n+1)!}}{\cancel{3} \cdot \cancel{3}} =$$

$$= \frac{3}{\infty} = 0 < 1 \Rightarrow \text{сходимость}$$

3. Различительный и интегральный признаки Коши

3.1. $\sum_{n=1}^{\infty} \left(\frac{2^n}{3n+1} \right)^n$

$$a_n = \left(\frac{2^n}{3n+1} \right)^n$$

$$L = \lim_{n \rightarrow \infty} \sqrt[n]{\left(\frac{2^n}{3n+1} \right)^n} = \frac{2^n}{3n+1} = \frac{2}{3 + \frac{1}{n}} = \frac{2}{3} < 1 \Rightarrow$$

$\Rightarrow$ **сходящееся**

3.2. $\sum_{n=1}^{\infty} \frac{1}{2^n} \cdot \left(\frac{n+1}{n} \right)^{n^2}$

$$a_n = \frac{1}{2^n} \cdot \left(\frac{n+1}{n} \right)^{n^2}$$

$$\lim_{n \rightarrow \infty} \sqrt[n]{\frac{1}{2^n} \cdot \left(\frac{n+1}{n} \right)^{n^2}} = \lim_{n \rightarrow \infty} \frac{1}{2} \cdot \left(\frac{n+1}{n} \right)^n = \lim_{n \rightarrow \infty} \frac{1}{2} \cdot \left(1 + \frac{1}{n} \right)^n =$$

$$= \frac{1}{2} \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n} \right)^{n \cdot \frac{1}{n} \cdot n} = \frac{1}{2} \lim_{n \rightarrow \infty} e = \frac{e}{2} > 1 \Rightarrow$$

$\Rightarrow$ **расходящееся**

3.3. $\sum_{n=1}^{\infty} \frac{1}{(n+1) \ln^3(n+1)}$

$$\int_1^{+\infty} \frac{dx}{(x+1) \ln^3(x+1)} dx =$$

$$\left[\begin{array}{l} \ln(x+1) = t \\ dt = \frac{1}{x+1} dx \end{array} \right] =$$

$$f(x) = \frac{1}{(x+1) \cdot \ln^3(x+1)} - \text{убывает}$$

$$= \int_1^{+\infty} \frac{1}{t^3} dt = -\frac{1}{2t^2} = -\frac{1}{2u^2(x+1)} \Big|_1^{+\infty} =$$

$$= -0 + \frac{1}{2u^2(2)} = \text{сходится}$$

$$3.4. \sum_{n=1}^{\infty} \frac{e^{-\sqrt{n}}}{\sqrt{n}}$$

Рассмотрим ф-у $f(x) = \frac{e^{-\sqrt{x}}}{\sqrt{x}}$ - убывает

$$\int_1^{+\infty} \frac{e^{-\sqrt{x}}}{\sqrt{x}} dx = \left[\begin{array}{l} \sqrt{x} = t \\ dt = \frac{1}{2\sqrt{x}} dx \\ 2 dt = \frac{1}{\sqrt{x}} dx \end{array} \right] =$$

$$= 2 \int_1^{+\infty} e^{-t} dt = 2 \left(e^{-\infty} + \frac{1}{e} \right) = \frac{2}{e} < 1 \Rightarrow \text{сходится}$$

4. Знаменательные числовые ряды.
Критерий Лейбница

$$4.1. \sum_{n=1}^{\infty} \frac{(-1)^n n}{n^2+1}$$

1) Проверим сходимость по Лейбницу (на убывание)
 $a_n = \frac{n}{n^2+1}$ (без знака)

$$a_1 = \frac{1}{2}; a_2 = \frac{2}{5}; a_3 = \frac{3}{10}; a_4 = \frac{4}{17}, \dots \Rightarrow$$

$\Rightarrow \{a_n\}$ - убывает

$$\lim_{n \rightarrow \infty} \frac{n}{n^2+1} = \lim_{n \rightarrow \infty} \frac{n \cdot \frac{1}{n}}{n^2 \cdot (1 + \frac{1}{n})} = 0 \Rightarrow \text{ряд сходится}$$

2) Проверим абсолютную сходимость:

$$a_n = \frac{n}{n^2+1} \sim \frac{1}{n} = b_n$$

$$\sum_{n=1}^{\infty} b_n = \sum_{n=1}^{\infty} \frac{1}{n} - \text{расходящееся}$$

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \lim_{n \rightarrow \infty} \frac{\frac{n}{n^2+1}}{\frac{1}{n}} = \lim_{n \rightarrow \infty} \frac{n^2}{n^2+1} = \lim_{n \rightarrow \infty} \left(1 - \frac{1}{n^2+1}\right) =$$

$= 1 \Rightarrow$ признак Коши и расходящееся

$\Rightarrow$ ряд расходящийся по признаку Лейбница
(условная сходимость)

4.2 $\sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt{n^2+2n}}$

1) Признак Лейбница

$$a_n = \frac{1}{\sqrt{n^2+2n}}$$

$$a_1 = \frac{1}{\sqrt{3}}; a_2 = \frac{1}{2}; \dots \Rightarrow \{a_n\} - \text{убывает}$$

$$\lim_{n \rightarrow \infty} \frac{1}{\sqrt{n^2+2n}} = 0 \Rightarrow \text{ряд сходится по кр. Лейбница}$$

2) Проверим абсолютную сходимость

$$a_n = \frac{1}{\sqrt{n^2+2n}} \sim \frac{1}{n} = b_n$$

$$\sum_{n=1}^{\infty} b_n = \sum_{n=1}^{\infty} \frac{1}{n} - \text{расходящееся}$$

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \lim_{n \rightarrow \infty} \frac{1}{\sqrt{n^2 + 2n}} \cdot \frac{n}{1} = \lim_{n \rightarrow \infty} \frac{n}{n \sqrt{1 + \frac{2}{n}}} =$$

$$= \frac{1}{1} = 1 \Rightarrow \text{расходимость}$$

$\Rightarrow$ условие сходимости по кр. Лейбница

4.3.
$$\sum_{n=1}^{\infty} \frac{(-1)^n n^2}{(n+1)!}$$

1) Признак Д'Аламбера

$$a_n = \frac{n^2}{(n+1)!}$$

$$a_{n+1} = \frac{(n+1)^2}{(n+2)!}$$

$$L = \lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \lim_{n \rightarrow \infty} \frac{(n+1)^2}{(n+2)!} \cdot \frac{(n+1)!}{n^2} =$$

$$= \lim_{n \rightarrow \infty} \frac{(n+1)^2 \cdot \cancel{(n+1)!}}{\cancel{(n+1)!} \cdot (n+2) \cdot n^2} = \lim_{n \rightarrow \infty} \frac{n^2 + 2n + 1}{n^3 + 2n^2} =$$

$$= \lim_{n \rightarrow \infty} \frac{n^3 \left(\frac{1}{n} + \frac{2}{n^2} + \frac{1}{n^3} \right)}{n^3 \left(1 + \frac{2}{n} \right)} = \frac{0}{1} = 0 < 1 \Rightarrow \text{сходим-}$$

ся абсолютно, можно проверить условие

сходимости нем.

4.4. $\sum_{n=1}^{\infty} \frac{(-1)^n n}{(n+1)^{1,5}}$ - упр. сходя.

1) Проверим абсолютную сходимость

$$a_n = \frac{n}{(n+1)^{1,5}} \sim \frac{n}{n^{1,5}} = \frac{1}{n^{0,5}} = \frac{1}{\sqrt{n}}$$

$$\sum_{n=1}^{\infty} b_n = \sum_{n=1}^{\infty} \frac{1}{\sqrt{n}} - \text{расходится } (p \leq 1)$$

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \lim_{n \rightarrow \infty} \frac{n}{\sqrt{(n+1)^3}} \cdot \frac{\sqrt{n}}{1} = \lim_{n \rightarrow \infty} \frac{n^{\frac{3}{2}}}{(n+1)^{\frac{3}{2}}} =$$

$$= \lim_{n \rightarrow \infty} \left(\frac{n}{n+1} \right)^{1,5} = \lim_{n \rightarrow \infty} \left(\frac{1}{1 + \frac{1}{n}} \right)^{1,5} = \left(\frac{1}{1+0} \right)^{1,5} = 1 \Rightarrow$$

$\Rightarrow$ расходится

2) Проверим условную сходимость

$$a_n = \frac{n}{(n+1)^{\frac{3}{2}}}$$

$$a_1 = \frac{1}{\sqrt{8}} ; a_2 = \frac{2}{\sqrt{27}} ; \dots \Rightarrow \{a_n\} - \text{убывающее}$$

$$\lim_{n \rightarrow \infty} \frac{n}{(n+1)^{\frac{3}{2}}} = \lim_{n \rightarrow \infty} \frac{1}{\frac{3}{2} \cdot \sqrt{n+1}} = \frac{1}{\infty} = 0 \Rightarrow \text{маж. сходящееся}$$

условно.

Функциональные ряды.

1. Исследовать на сходимость степенный ряд

1.1.
$$\sum_{n=1}^{\infty} \frac{(n+1)(x+2)^n}{n^2-2n+7}$$

$$\lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} = \lim_{n \rightarrow \infty} \frac{(n+2)(x+2)^{n+1}}{(n+1)^2-2(n+1)+7} \cdot \frac{n^2-2n+7}{(n+1)(x+2)^n} =$$

$$= |x+2| \lim_{n \rightarrow \infty} \frac{(n+2)(n^2-2n+7)}{((n+1)^2-2(n+1)+7)(n+1)} = |x+2| \lim_{n \rightarrow \infty} \frac{n^3}{n^3} = |x+2| < 1$$

$$-1 < x+2 < 1$$

$$-3 < x < -1$$

Гранича $x = -3$:

$$\sum_{n=1}^{\infty} \frac{(n+1)(-1)^n}{n^2-2n+7} = \sum_{n=1}^{\infty} (-1)^n \cdot \frac{n+1}{n^2-2n+7}$$

$$a_n = \frac{n+1}{n^2-2n+7}$$

По формуле Лейбница для знакоперемен. рядов

$$1) a_1 = \frac{2}{6} = \frac{1}{3}; a_2 = \frac{3}{7}; a_3 = \frac{4}{10} = \frac{2}{5}; a_4 = \frac{5}{15} = \frac{1}{3}, \dots$$

$\{a_n\}$ убывает с $n_0 = 2$

$$2) \lim_{n \rightarrow \infty} \frac{n+1}{n^2-2n+7} = \lim_{n \rightarrow \infty} \frac{n}{n^2} = \lim_{n \rightarrow \infty} \frac{1}{n} = 0 \Rightarrow \text{ряд сходится}$$

Проверка $x = -1$:

$$\sum_{n=1}^{\infty} \frac{n+1}{n^2-2n+7}$$

$$a_n = \frac{n+1}{n^2-2n+7} \sim \frac{n}{n^2} = \frac{1}{n} = b_n$$

$$\sum_{n=1}^{\infty} b_n = \sum_{n=1}^{\infty} \frac{1}{n} - \text{расходящееся}$$

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \lim_{n \rightarrow \infty} \frac{n+1}{n^2-2n+7} \cdot \frac{n}{1} = \lim_{n \rightarrow \infty} \frac{n^2}{n^2} = 1 \neq 0$$

Ряд сходящееся при $-3 \leq x < -1$

1.2.
$$\sum_{n=1}^{\infty} \frac{(x-1)^n}{2^n (n^2-3n+10)}$$

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \left| \frac{(x-1)^{n+1}}{2^{n+1} \cdot ((n+1)^2 - 3(n+1) + 10)} \cdot \frac{2^n (n^2 - 3n + 10)}{(x-1)^n} \right| =$$

$$= |x-1| \lim_{n \rightarrow \infty} \frac{1}{2} \cdot \frac{n^2}{n^2} = |x-1| \cdot \frac{1}{2} < 1$$

$$|x-1| < 2; \quad -2 < x-1 < 2; \quad -1 < x < 3$$

Проверка $x = -1$:

$$\sum_{n=1}^{\infty} \frac{(-2)^n}{2^n (n^2-3n+10)} = \sum_{n=1}^{\infty} (-1)^n \cdot \frac{1}{n^2-3n+10}; \quad a_n = \frac{1}{n^2-3n+10}$$

$$1) a_1 = \frac{1}{8}; a_2 = \frac{1}{8}; a_3 = \frac{1}{10}; a_4 = \frac{1}{14}, \dots \Rightarrow \{a_n\} -$$

убывающим $c_n = 2$

$$2) \lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \frac{1}{n^2 - 3n + 10} = \lim_{n \rightarrow \infty} \frac{1}{n^2} = 0 \Rightarrow \text{сходимость по}$$

пр. Лебнера

$$x = 3: \sum_{n=1}^{\infty} \frac{2^n}{2^n(n^2 - 3n + 10)} = \sum_{n=1}^{\infty} \frac{1}{n^2 - 3n + 10}$$

$$a_n = \frac{1}{n^2 - 3n + 10} \sim \frac{1}{n^2} = b_n - \text{сходимость}$$

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \frac{1}{n^2 - 3n + 10} \cdot \frac{n^2}{1} = \lim_{n \rightarrow \infty} \frac{n^2}{n^2} = 1$$

$$\Rightarrow -1 \leq x \leq 3$$

$$1.3. \sum_{n=1}^{\infty} \frac{(x+3)^n \cdot n \cdot (-1)^{n+1}}{3^{n+1} \cdot \sqrt{4n^3+2}}$$

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \left| \frac{(x+3)^{n+1} \cdot (n+1)}{3^{n+2} \sqrt{4(n+1)^3+2}} \cdot \frac{3^{n+1} \cdot \sqrt{4n^3+2}}{(x+3)^n \cdot n \cdot (-1)^{n+1}} \right| =$$

$$= |x+3| \lim_{n \rightarrow \infty} \frac{|n+1| \sqrt{4n^3+2}}{3 \sqrt{4(n+1)^3+2} \cdot n} = |x+3| \cdot \lim_{n \rightarrow \infty} \frac{n^{2,5}}{3n^{2,5}} = \frac{|x+3|}{3} < 1$$

$$-1 < \frac{|x+3|}{3} < 1$$

$$-3 < |x+3| < 3$$

$$-6 < x < 0$$

$$\underline{x = -6:}$$

$$a_n = \frac{(-3)^n \cdot n \cdot (-1)^{n+1}}{3^{n+1} \cdot \sqrt{4n^3+2}} = \sum_{n=1}^{\infty} \frac{3^n \cdot n \cdot (-1)^{n+1}}{3^{n+1} \sqrt{4n^3+2}} = - \sum_{n=1}^{\infty} \frac{n}{3\sqrt{4n^3}}$$

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \frac{n}{3\sqrt[3]{4n^3+1}} = \lim_{n \rightarrow \infty} \frac{n}{6n^3} = \lim_{n \rightarrow \infty} \frac{1}{6n^2} = 0 \text{ — монотонно}$$

$$x=0:$$

$$\sum_{n=1}^{\infty} \frac{3^n \cdot n \cdot (-1)^{n+1}}{3^{n+1} \sqrt[3]{4n^3+2}} = \frac{1}{3} \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{\sqrt[3]{4n^3+2}} \Rightarrow a_n = \frac{1}{\sqrt[3]{4n^3+2}}$$

$$a_1 = \frac{1}{\sqrt[3]{7}}, a_2 = \frac{1}{\sqrt[3]{35}}, \dots \Rightarrow \{a_n\} \text{ убывает.}$$

$$\lim_{n \rightarrow \infty} \frac{1}{\sqrt[3]{4n^3+2}} = 0 \Rightarrow \text{сходимость}$$

$$-6 \leq x \leq 0$$

$$1.4. \sum_{n=1}^{\infty} \frac{x^{2n} \cdot (n+2)}{6n^3+5n+4}$$

$$\lim_{n \rightarrow \infty} \left| \frac{u_{n+1}}{u_n} \right| = \lim_{n \rightarrow \infty} \left| \frac{\cancel{x^{2n+1}} \cdot (n+3)}{6(n+1)^3+5(n+1)+4} \cdot \frac{6n^3+5n+4}{\cancel{x^{2n}} \cdot (n+2)} \right| =$$

$$= x^2 \lim_{n \rightarrow \infty} 1 = x^2 < 1$$

$$x^2 < 1$$

$$-1 < x < 1$$

$$x=-1:$$

$$\sum_{n=1}^{\infty} \frac{(-1)^{2n} \cdot (n+2)}{6n^3+5n+3} = \sum_{n=1}^{\infty} \frac{(n+2)}{6n^3+5n+3}$$

$$a_n = \frac{n+2}{6n^3+5n+3} \sim \frac{n}{n^3} = \frac{1}{n^2} = b_n$$

$$\sum_{n=1}^{\infty} b_n = \sum_{n=1}^{\infty} \frac{1}{n^2} - \text{сходящееся}$$

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \lim_{n \rightarrow \infty} \frac{n+2}{6n^3+5n+3} \cdot \frac{n^2}{1} = 1 \Rightarrow -1 \leq x \leq 1$$

2. Разложить по степеням x .

2.1. $y = x^2 \cdot e^{-x} \rightarrow e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$

$$e^{-x} = 1 - \frac{x}{1!} + \frac{x^2}{2!} - \frac{x^3}{3!} + \dots$$

$$x^2 \cdot e^{-x} = x^2 - \frac{x^3}{1!} + \frac{x^4}{2!} - \frac{x^5}{3!} = \sum_{n=0}^{\infty} (-1)^n \cdot \frac{x^{n+2}}{n!}$$

2.2. $y = \cos(x^3) \rightarrow \cos(x) = 1 - \frac{1}{2!} x^2 + \frac{1}{4!} x^4 - \frac{1}{6!} x^6 + \dots$

$$\cos(x^3) = 1 - \frac{1}{2!} x^6 + \frac{1}{4!} x^{12} - \frac{1}{6!} x^{18} + \dots$$

2.3. $y = \ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} + \dots$

$$\ln(1+2x) = 2x - \frac{(2x)^2}{2} + \frac{(2x)^3}{3} - \dots$$

2.4. $\frac{1}{1-x} = 1 + x + x^2 + x^3 + \dots$

$$\frac{1}{1+x^2} = 1 - x^2 + x^4 - x^6 + \dots$$

$$\operatorname{arctg}(x) = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \dots \quad (\text{взято из таблицы})$$

3. Вычислить с точностью $\varepsilon = 10^{-3}$.

3.1. $\int_0^1 e^{-x^3} dx$

$$e^{-x^3} = 1 - \frac{x^3}{1!} + \frac{x^6}{2!} - \frac{x^9}{3!} + \frac{x^{12}}{4!} - \frac{x^{15}}{5!} + \dots$$

$$\int_0^1 e^{-x^3} dx = \int_0^1 \left(1 - \frac{x^3}{1!} + \frac{x^6}{2!} - \frac{x^9}{3!} + \dots \right) dx =$$

$$= \left(x - \frac{x^4}{4} + \frac{x^7}{7} - \frac{x^{10}}{10} + \dots \right) \Big|_0^1 = \boxed{1 - \frac{1}{4} + \frac{1}{7} - \frac{1}{10} + \dots}$$

$$+ \frac{1}{312} - \frac{1}{16 \cdot 120} < 10^{-3} = 0,783$$

3.2. $\int_0^1 \frac{\sin x}{x} dx$

$$\sin(x) = \frac{x}{1} - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots$$

$$\frac{\sin(x)}{x} = 1 - \frac{x^2}{3!} + \frac{x^4}{5!} - \dots$$

$$\int_0^1 \frac{\sin(x)}{x} dx = \int_0^1 \left(1 - \frac{x^2}{3!} + \frac{x^4}{5!} - \dots \right) dx = \left(x - \frac{x^3}{18} + \frac{x^5}{600} - \frac{x^7}{7 \cdot 7!} + \dots \right)$$

$$= 1 - \frac{1}{18} + \frac{1}{600} = 0,946$$

3.3. $\int_0^{0,4} \frac{\ln(1+x^2)}{x} dx$

$$\ln(1+x) = x - \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

$$\ln(1+x^2) = x^2 - \frac{x^4}{2!} + \frac{x^6}{3!} + \dots$$

$$\frac{\ln(1+x^2)}{x} = x - \frac{x^3}{2!} + \frac{x^5}{3!} + \dots$$

$$\int_0^{0,4} \left(x - \frac{x^3}{2!} + \frac{x^5}{3!} - \dots \right) dx = \frac{x^2}{2} - \frac{x^4}{8} + \left(\frac{x^6}{36} \right) \Big|_0^{0,4} =$$

$\uparrow$
 10^{-3}

$$= 0,08 - 0,0032 = 0,0768$$

3.4. $\int_0^1 \cos(x^3) dx$

$$\cos(x) = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots$$

$$\cos(x^3) = 1 - \frac{x^6}{2!} + \frac{x^{12}}{4!} - \frac{x^{18}}{6!} + \dots$$

$$\int_0^1 \cos(x^3) dx = \int_0^1 \left(1 - \frac{x^6}{2!} + \frac{x^{12}}{4!} - \frac{x^{18}}{6!} + \dots \right) dx =$$

$$= x - \frac{x^7}{14} + \frac{x^{13}}{312} - \left(\frac{x^{19}}{13680} + \dots \right) \Big|_0^1 =$$

$\uparrow$
 $< 10^{-3}$

$$= 1 - \frac{1}{14} + \frac{1}{312} = 0,932$$

Разложение в КР

Пример. Разложим в ряд Маклорена

$$f(x) = \ln(x+2) = \ln\left(2\left(\frac{x}{2} + 1\right)\right) = \ln 2 + \ln\left(1 + \frac{x}{2}\right) = \\ = \ln 2 + \frac{x}{2} - \frac{\left(\frac{x}{2}\right)^2}{2} + \frac{\left(\frac{x}{2}\right)^3}{3} \dots$$

Область сходимости $-1 < \frac{x}{2} \leq 1$
 $-2 < x \leq 2$

$$\ln 2 = \ln(1+1) = 1 - \frac{1^2}{2} + \frac{1^3}{3} - \frac{1^4}{4} + \dots$$

Пример. $f(x) = \frac{2}{3-x} = \frac{2}{3(1-\frac{x}{3})} =$
 $= \frac{2}{3} \cdot \frac{1}{1-\frac{x}{3}} = \frac{2}{3} \left(1 + \frac{x}{3} + \frac{x^2}{9} + \frac{x^3}{27} + \dots\right)$ $-1 < \frac{x}{3} < 1$
 $-3 < x < 3$

Пример. Разложим $\sin(1)$ с точностью до 0,001

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!}$$

$$\sin 1 = 1 - \frac{1}{6} + \frac{1}{120} - \left(\frac{1}{5040}\right) + \dots$$

$\approx 10^{-3}$

$$\sin 1 \approx 0,843 \pm 0,001$$

Пример. $y'' = x^2 + y^2$, $y(-1)=2$, $y'(-1)=\frac{1}{2}$

$$y(x) = y(-1) + \frac{y'(-1)}{1!}(x+1) + \frac{y''(-1)}{2!}(x+1)^2 + \dots + \frac{y^{(n)}(-1)}{n!}(x+1)^n$$

$$y'' = x^2 + y^2$$

$$y''(-1) = (-1)^2 + 2^2 = 5$$

$$y''' = 2x + 2y \cdot y'$$

$$y'''(-1) = -2 + 2 \cdot 2 \cdot \frac{1}{2} = 0$$

$$y^{IV} = 2 + 2(y' \cdot y' + y \cdot y'')$$

$$y^{IV}(-1) = 2 + 2\left(\frac{1}{4} + 2 \cdot 5\right) = 22,5$$

$$y^V = 2(2 \cdot y' \cdot y'' + y' y'' + y y''') = 6y' y'' + 2y y'''$$

$$y^V(-1) = 6 \cdot \frac{1}{2} \cdot 5 + 2 + 2 \cdot 2 \cdot 0 = 15$$

Umwert: $y(x) = 2 + \frac{1}{2}(x+1) + \frac{5}{2}(x+1)^2 + \frac{22,5}{24}(x+1)^3 + \frac{15}{120}(x+1)^4$

Zagars 1. $\sum_{n=1}^{\infty} \frac{(n^2+3)^{\frac{2}{3}}}{(n+5)(n^2+3)}$

$$a_n = \frac{(n^2+3)^{\frac{2}{3}}}{(n+5)(3+n^2)} \sim \frac{n^{\frac{4}{3}}}{n^3} = \frac{1}{n^{\frac{5}{3}}} = b_n$$

$$\sum_{n=1}^{\infty} b_n = \sum_{n=1}^{\infty} \frac{1}{n^{\frac{5}{3}}} - \text{crescens (} p = \frac{5}{3} > 1 \text{)}$$

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \lim_{n \rightarrow \infty} \frac{(n^2+3)^{\frac{2}{3}}}{(n+5)(3+n^2)} \cdot \frac{n^{\frac{5}{3}}}{1} = \lim_{n \rightarrow \infty} \frac{n^{\frac{4}{3}} \cdot n^{\frac{5}{3}}}{n \cdot n^2} =$$

$$= \lim_{n \rightarrow \infty} \frac{n^3}{n^3} = 1 \neq 0$$

По предельному признаку ср. a_n -сходится

Задача 2

$$\sum_{n=1}^{\infty} \frac{3^{n-1}}{(2n+1)!}$$

$$a_n = \frac{3^{n-1}}{(2n+1)!}$$

$$a_{n+1} = \frac{3^n}{(2n+3)!}$$

$$\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \lim_{n \rightarrow \infty} \frac{3^n \cdot (2n+1)!}{(2n+3)! \cdot 3^{n-1}} = 3 \lim_{n \rightarrow \infty} \frac{1 \cdot 2 \cdot 3 \cdot \dots \cdot (2n+1)}{1 \cdot 2 \cdot 3 \cdot \dots \cdot (2n+1)(2n+2)(2n+3)} =$$

$$= 3 \lim_{n \rightarrow \infty} \frac{1}{(2n+2)(2n+3)} = 0 < 1$$

по признаку Д'Аламбера $\sum_{n=1}^{\infty} a_n$ сходится

Задача 3

$$\sum_{n=1}^{\infty} 2^n \cdot \left(\frac{4n-1}{7n+3} \right)^{2n}$$

$$a_n = 2^n \cdot \left(\frac{4n-1}{7n+3} \right)^{2n}$$

$$\lim_{n \rightarrow \infty} \sqrt[n]{a_n} = \lim_{n \rightarrow \infty} 2 \cdot \left(\frac{4n-1}{7n+3} \right)^2 = 2 \cdot \left(\frac{4}{7} \right)^2 = \frac{32}{49} < 1$$

по предельному признаку $\sum_{n=1}^{\infty} a_n$ сходится.

Задача 4.

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1} (3+n)}{n^3 + 4n - 1}$$

$$a_n = \frac{3+n}{n^3+4n-1}$$

$$1) a_1 = \frac{4}{4} = 1; a_2 = \frac{5}{15} = \frac{1}{3}; a_3 = \frac{6}{38} = \frac{3}{19}; \dots$$

$\Rightarrow \{a_n\}$ - убывает

$$2) \lim_{n \rightarrow \infty} \frac{3+n}{n^3+4n-1} = \lim_{n \rightarrow \infty} \frac{n}{n^3} = \lim_{n \rightarrow \infty} \frac{1}{n^2} = 0$$

$\Rightarrow$ По признаку Лейбница сходится

Рассмотрим $\sum_{n=1}^{\infty} \frac{3+n}{n^3+4n-1}$

$$b_n = \frac{1}{n^2} - \text{сходится}$$

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \lim_{n \rightarrow \infty} \frac{3+n}{n^3+4n-1} \cdot \frac{n^2}{1} = 1 \neq 0$$

$$\sum_{n=1}^{\infty} a_n - \text{сходится}$$

Умнож: $\sum_{n=1}^{\infty} (-1)^{n+1} a_n$ - сходится абсолютно.

Задача 5. $\sum_{n=1}^{\infty} \frac{(x+4)^n}{4^n \sqrt{n^2+5}}$

$$\lim_{n \rightarrow \infty} \left| \frac{(x+4)^{n+1}}{4^{n+1} \sqrt{(n+1)^2+5}} \cdot \frac{4^n \sqrt{n^2+5}}{(x+4)^n} \right| = \frac{|x+4|}{4} \lim_{n \rightarrow \infty} \frac{\sqrt{n^2+5}}{\sqrt{(n+1)^2+5}} =$$

$$= \frac{|x+4|}{4} < 1$$

$$\begin{aligned} |x+4| &< 4 \\ -4 &< x+4 < 4 \\ -8 &< x < 0 \end{aligned}$$

$$x=0: \sum_{n=1}^{\infty} \frac{4^n}{4^n \sqrt{n^2+5}} = \sum_{n=1}^{\infty} \frac{1}{\sqrt{n^2+5}}$$

$$a_n = \frac{1}{\sqrt{n^2+5}} \sim \frac{1}{n} = b_n - \text{расходящиеся}$$

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = 1 \Rightarrow \sum_{n=1}^{\infty} a_n - \text{расходящиеся}$$

$$x = -8:$$

$$\sum_{n=1}^{\infty} \frac{(-4)^n}{4^n \sqrt{n^2+5}} = \sum_{n=1}^{\infty} (-1)^n \cdot \frac{1}{\sqrt{n^2+5}}$$

$$1) a_1 = \frac{1}{\sqrt{6}}; a_2 = \frac{1}{\sqrt{9}}; a_3 = \frac{1}{\sqrt{14}}, \dots - \text{убываем}$$

$$2) \lim_{n \rightarrow \infty} \frac{1}{\sqrt{n^2+5}} = 0 \Rightarrow \text{сходящиеся}$$

Уточн: Ряд сходящиеся при $x \in [-8; 0)$

Задача 6.

$$\int_0^1 \sqrt{1+x^2} dx, \quad \epsilon = 0,001$$

$$(1+x)^{\frac{1}{2}} = 1 + \frac{1}{2}x - \frac{1}{8}x^2 + \frac{1}{16}x^3 - \frac{5}{128}x^4 + \dots$$

$$(1+x^2)^{\frac{1}{2}} = 1 + \frac{1}{2}x^2 - \frac{1}{8}x^4 + \frac{1}{16}x^6 - \frac{5}{128}x^8 + \dots$$

$$\int_0^1 \left(1 + \frac{1}{2}x^2 - \frac{1}{8}x^4 + \frac{1}{16}x^6 - \frac{5}{128}x^8 + \dots \right) dx =$$

$$= \left(x + \frac{x^3}{6} - \frac{x^5}{40} + \frac{x^7}{112} - \frac{5x^9}{1152} + \dots \right) \Big|_0^1 = 1 + \frac{1}{6} - \frac{1}{40} + \frac{1}{112} - \frac{5}{1152} =$$

$$= 1,146$$